

Potential Interactions between ACI 544 and ACI 318 FRC as a Sustainable Infrastructure Material

Barzin Mobasher

Professor

School of Sustainable Engineering and the Built Environment

Ira A. Fulton Schools of Engineering

Arizona State University

Tempe, AZ 85287-5306

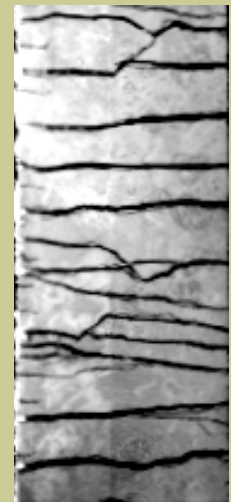
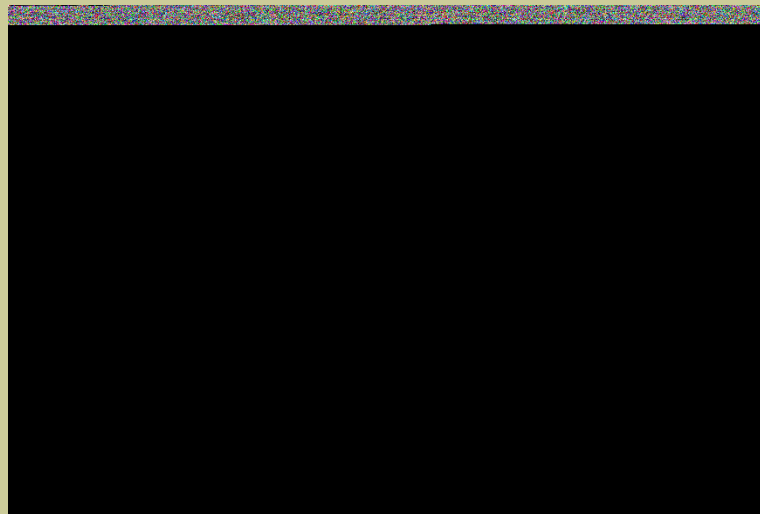
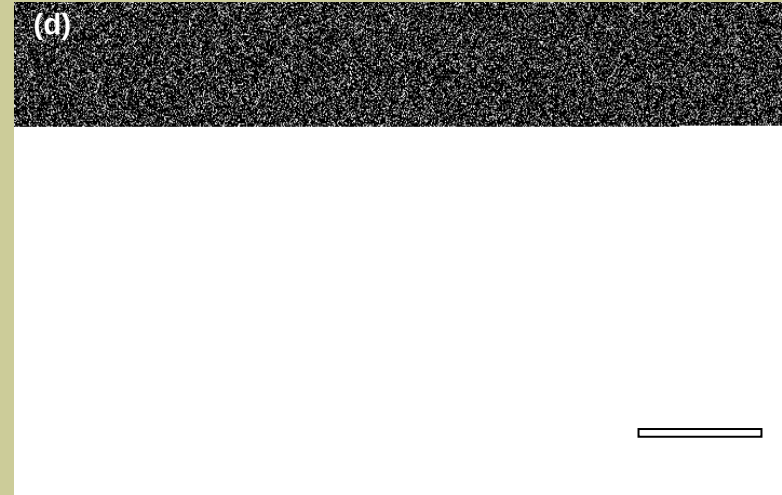


Opportunities in use of FRC in Structural Applications

- The phenomenal growth of the market for fiber reinforced concrete market is a key motivator for addressing sustainability
- Economy, labor, time, materials characteristics and performance
- ACI 544 has played a significant role in developing documents to showcase the performance of FRC materials
- Design opportunities:
 - Ductility, durability, crack width, stiffness, cracked section modulus.
 - Shear
 - A hybrid approach of combining reinforcement and fibers is the key to addressing sustainability
 - Minimum reinforcement requirements.

Materials characteristics

- Ductility
- Toughening
- Improved tensile strength
- Increase level of energy absorption
- Fatigue life, impact/explosive loading
- Seismic resistance
- Steel work, labor, construction time.
- Corrosion damage
- Long-term repair and maintenance.

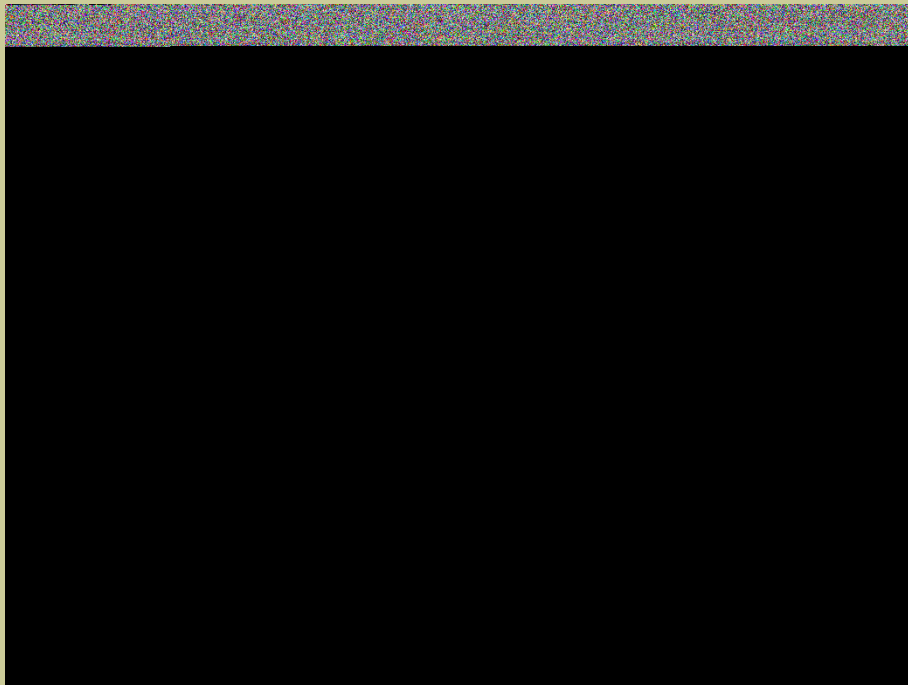


Seismic Design, Coupling beams, ACI-318 provisions

Eliminate a substantial amount of reinforcing bars by using a highly flowable, steel fiber-reinforced concrete where the fibers are introduced during the concrete mixing process.



Canal lining, WWF, or rebar replacement



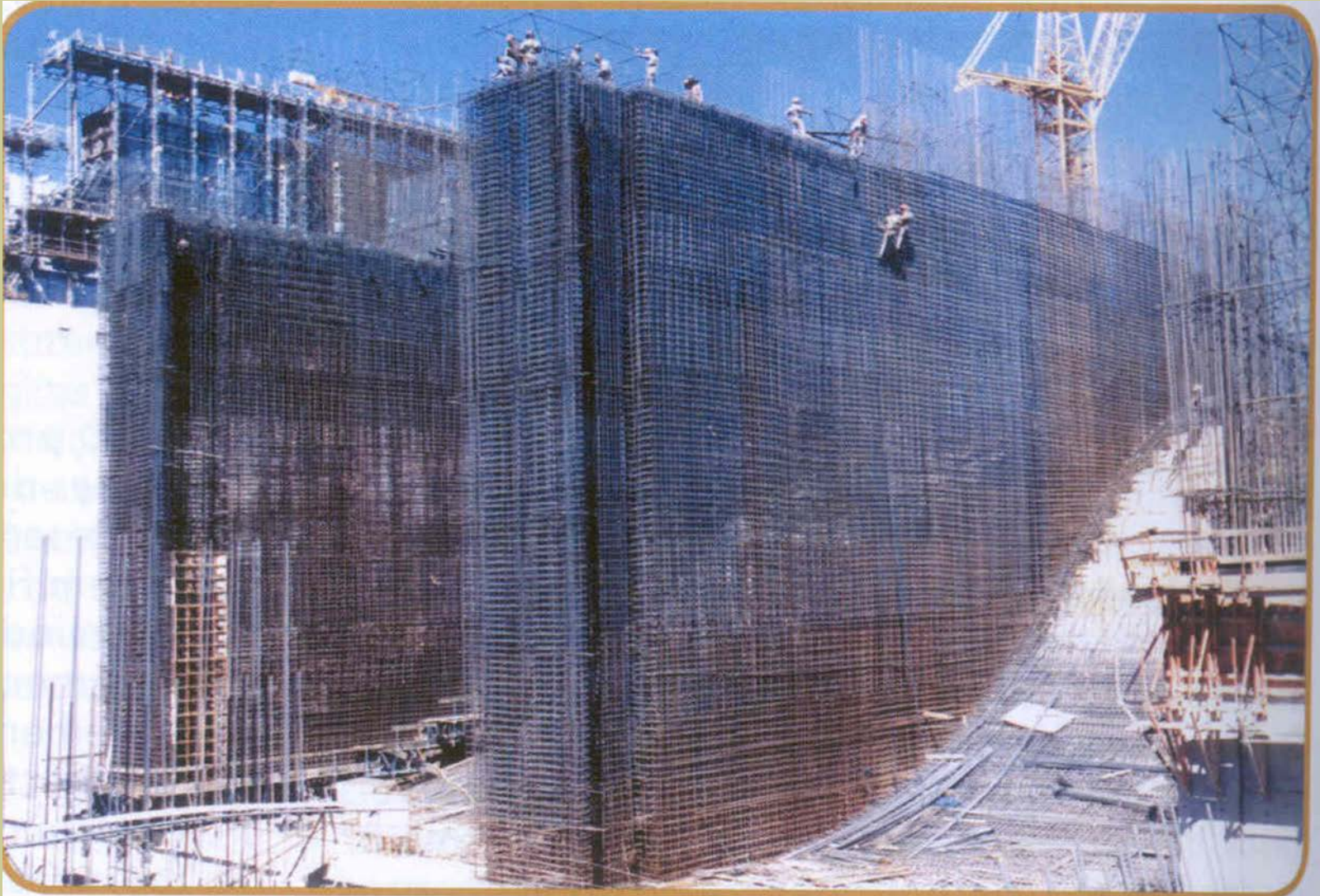
Traditional #5 rebar layout

Photo Courtesy: Rick Shelly, Pulice Construction

Fiber Reinforced Concrete Mix

Photo Courtesy: Pima-Maricopa Irrigation Project, Sacaton, Arizona

Spillway: high density of REBARs



Precast Tunnel lining for Rio de Janeiro Metro Line #4



Tunnel Lining

- Fiber-reinforced concrete can be used in **tunnel lining segments**.
- Macro steel fibers and macro synthetic fibers are mainly used for this type of application.
- Using fibers can reduce or even eliminate the steel rebar reinforcement which results in faster and cheaper production.



M. Moccichino et al., 2010 world tunnel conference, Italy

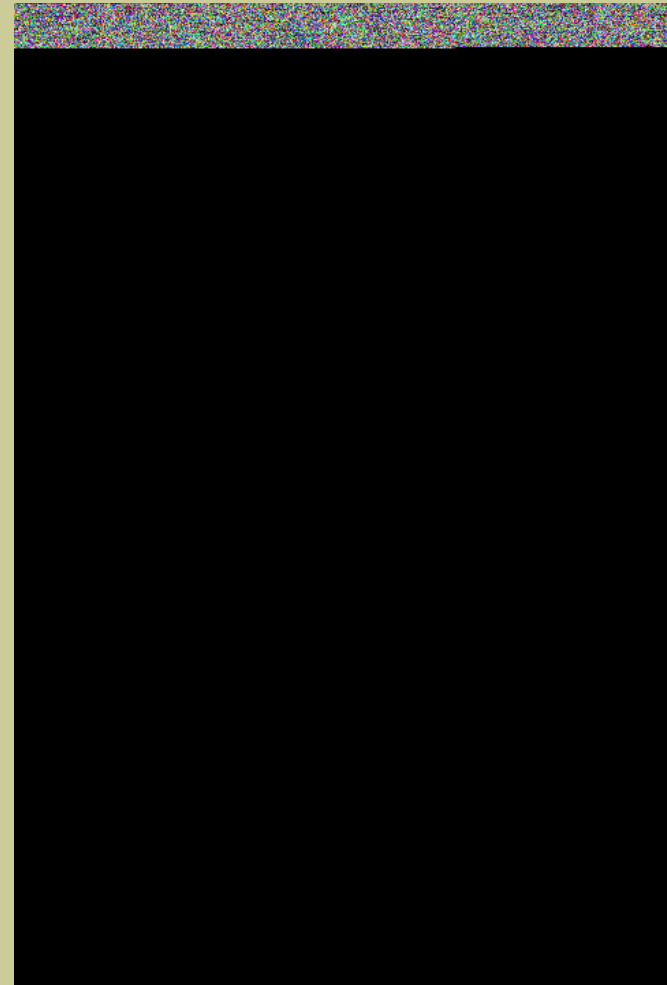


BEKAERT

FRC Engineering Applications

Ductility, Durability, Sustainability

- The type and volume fraction of fibers affect the level of energy absorption
- increased energy absorption, fatigue life, impact/explosive loading conditions, and seismic resistance
- Pavements/slabs, Precast components, Shotcrete



Structural Design with FRC Materials: testing, modeling, analysis and Design

Design with Strain Softening materials

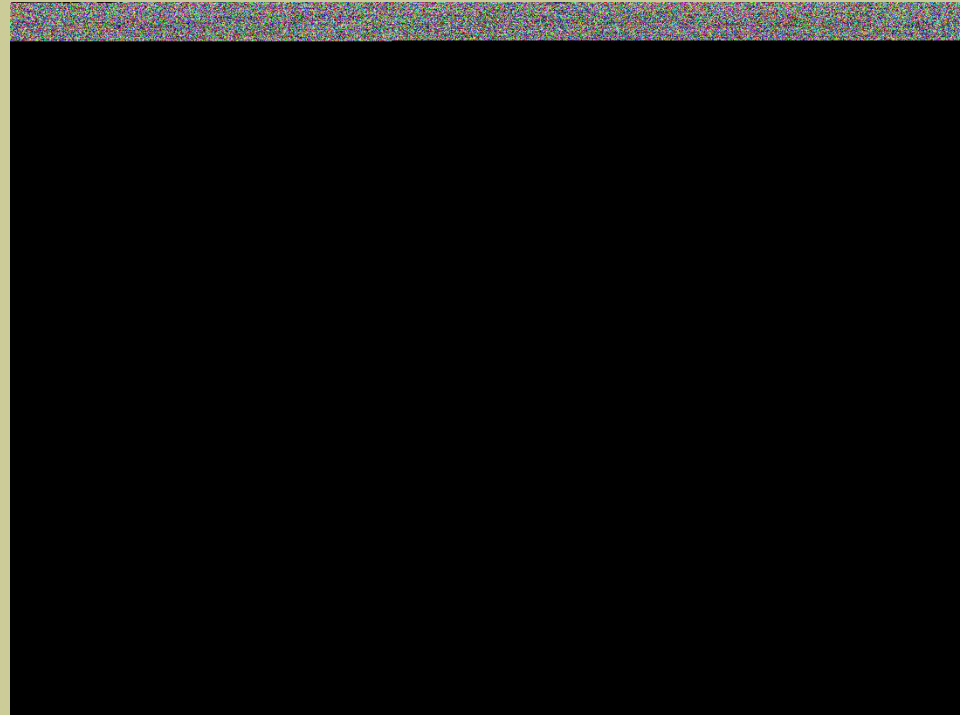
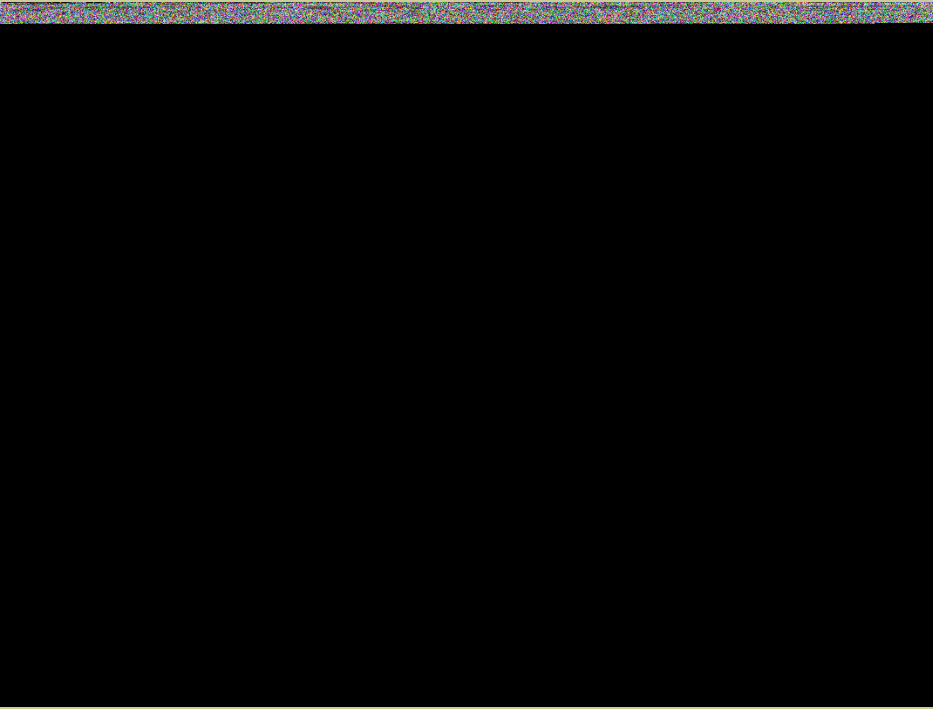


Elevated slabs

Precast panels

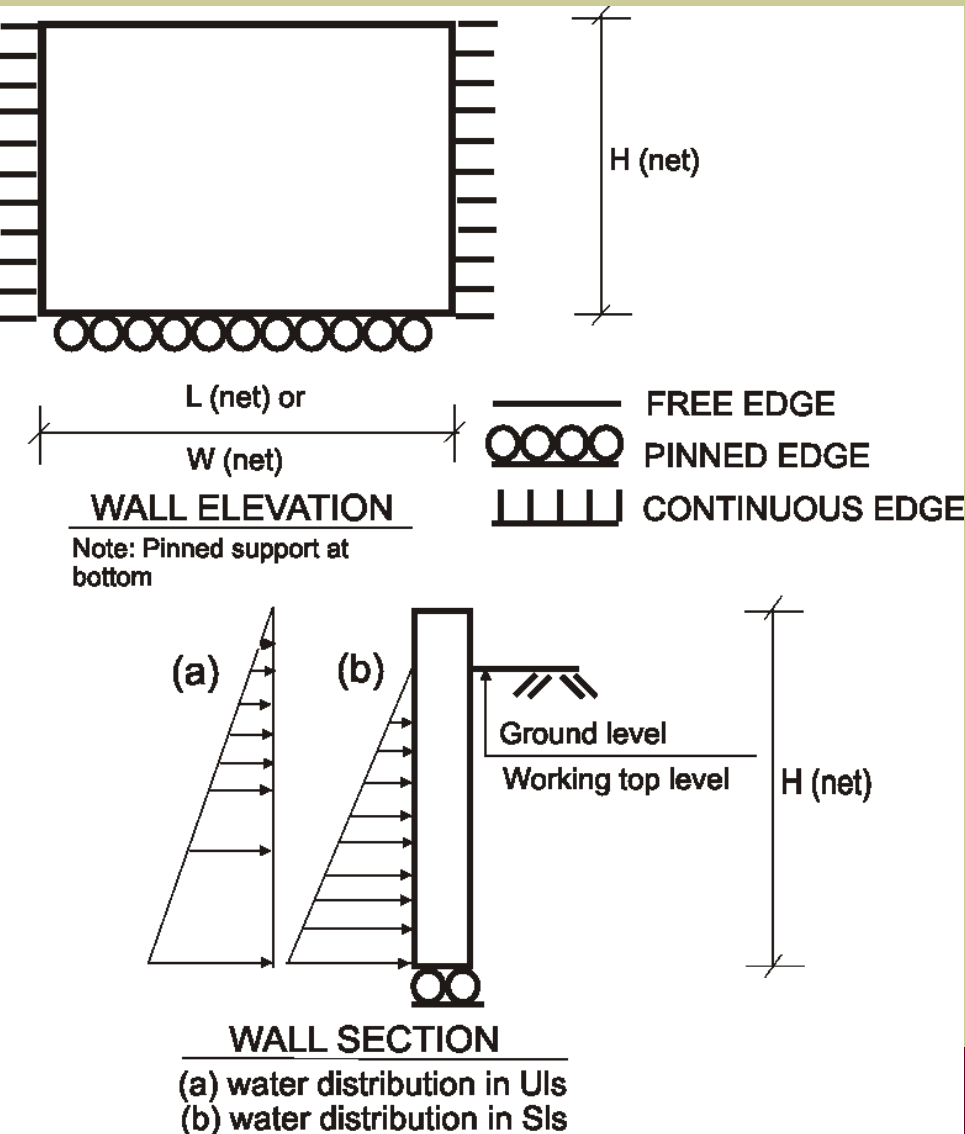
Shotcrete applications

Safety, mobilization, and Cost savings due to reduced section sizes



Analysis of Precast Wall Panels

- Assume continuous wall, pin connection at the bottom and free at the top
- Lateral water pressure in ultimate and serviceability limit states



Wind Turbine Foundation



ACI related International Committee Reports

ACI 544.5R-10 “Physical Properties and Durability of Fiber-Reinforced Concrete,” Report, ACI Committee, p. 31, (2010).

ACI 544.6R-15 Report on Design and Construction of Steel Fiber-Reinforced Concrete Elevated Slabs (2015)

ACI 544.7R-16 Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments (2016)

ACI 544.8R-16: Report on Indirect Method to Obtain Stress-Strain Response of Fiber-Reinforced Concrete (FRC), ACI Committee 544 ACI 544.8R (2016)

ACI 544.9R-17: Report on Measuring Mechanical Properties of Hardened Fiber-Reinforced Concrete, (2017)

https://www.concrete.org/Portals/0/Files/PDF/Previews/544.9R-17_preview.pdf

ACI 544.10R-17: Report on Measuring Properties of Fresh Fiber-Reinforced Concrete, (2018)

ACI 544-4R ACI 544.4R-18: Report on Structural Design with Fiber-Reinforced Concrete (2018)

Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments

Reported by ACI Committee 544

Barzin Mobasher, Chair
Neven Krstulovic-Opara, Secretary

Clifford N. MacDonald, Membership Secretary

Corina-Maria Aldea
Emmanuel K. Atiogbe
Mehdi Bakhshi*
Nemkumar Banthia
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Sidney Freedman
Christian Meyer
Henry J. Molloy
Antoine E. Naaman
Venkataswamy Ramakrishnan

*Chair of the task group that prepared this report.

¹Individuals who prepared this report.

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ACI encourages the development and appropriate use of new and emerging technologies through the publication of the Emerging Technology Series. This series presents information and recommendations based on available test data, technical reports, limited experience with field applications, and the opinions of committee members. The presented information and recommendations, and their basis, may be less fully developed and tested than those for more mature technologies. This report identifies areas in which information is believed to be less fully developed, and describes research needs. The professional using this document should understand the limitations of this document and exercise judgment as to the appropriate application of this emerging technology.

Fiber reinforcement has emerged as an alternative to traditional reinforcing bars and welded wire mesh reinforcement for precast concrete tunnel segments. Due to significantly improved post-cracking behavior and crack control characteristics, fiber-reinforced concrete (FRC) segments offer advantages over traditional

reinforced concrete segments such as saving cost and reducing production time while developing a more robust product with improved handling and long-term durability. Specific guidance on the design of fiber-reinforced precast concrete tunnel segments is needed for this emerging technology. This document offers general information on the history of FRC precast segments from tunneling projects throughout the world; a procedure for structural analysis and design based on governing load cases; and a description of the material parameters, tests, and analyses required to complete the design. The proposed guidelines are based on the knowledge gained from experimental research, analytical work, and the experience gained on numerous FRC precast tunnel projects.

Keywords: crack widths; earth pressure; fibers; fiber-reinforced concrete; grout pressure; hydrostatic pressure; lining; precast segment; stripping; surcharge load; thrust jack forces; tunnel.

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1.1—Introduction, p. 2

1.2—Scope and limitations, p. 3

1.3—Applications and uses in existing tunnels, p. 3

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Report on Design and Construction of Steel Fiber-Reinforced Concrete Elevated Slabs

Reported by ACI Committee 544

Barzin Mobasher,*† Chair
Neven Krstulovic-Opara, Secretary

Clifford N. MacDonald, Membership Secretary

Corina-Maria Aldea
Nemkumar Banthia
Joaquim Oliveira Barros*
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Dirk E. Nemegeer
Michael J. Roach
Morris Schupack¹
Ake Skarendahl
Ben L. Tilsen
Eldon G. Tipping
Jean Francois Trotter

*Members who prepared this report.

†Chair of the task group who drafted this report.

¹Deceased.

Construction of slabs in areas with weak soil conditions has commonly used pile-supported slab structural design so that the adverse effects of soil-structure interaction in terms of differential settlement, cracking, or long-term serviceability problems are avoided. In this application, the construction of slabs on closely spaced pile caps (typical span-depth ratios between 8 and 30) is referred to as elevated ground slabs (EGSs). These slabs may be subjected to moderately high loading, such as concentrated point loading of up to 44 kip (150 kN) and uniformly distributed loadings of 1000 lb/ft² (50 kN/m²). The dynamic loadings may be due to moving loads such as forklifts, travel lifts, and other

material handling equipment. Fiber-reinforced concrete (FRC) has been successfully used to address the structural design of these slabs. Based on the knowledge gained, the area has been extended to a construction practice for slabs supported by columns as well. Applications are further extended to multi-story building applications. This report addresses the methodology for analysis, design, and construction of steel FRC (SFRC) slabs supported on piles or columns (also called elevated SFRC [E-SFRC]). Sections of the report address the history, practice, applications, material testing, full-scale testing, and certifications. By compiling the practice and knowledge in the analysis design with FRC materials, the steps in the design approach based on ultimate strength approach using two-way slab mechanisms are presented. The behavior of a two-way system may not require the flexural strength of conventional reinforced concrete (RC) because of redistribution, redundancy, and failure mechanisms.

Methods of construction, curing, and full-scale testing of slabs are also presented. A high dosage of deformed steel fibers (85 to 170

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Report on Measuring Mechanical Properties of Hardened Fiber-Reinforced Concrete

Reported by Committee 544

Barzin Mobasher*, Chair
Neven Krstulovic-Opara, Secretary

Clifford N. MacDonald*, Membership Secretary

Corina-Maria Aldea
Emmanuel K. Attigbo
Mehdi Bakshi
Nemkumar Bonthia
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P. N. Balaguru
Hiran Price Ball Jr.
Gordon B. BatsonArnon Bentur
Andrzej M. Brandt
James I. DanielSidney Freedman
Christian Meyer
Henry J. MolloyAntoine E. Naaman*
Venkataswamy Ramakrishnan

*Members of subcommittee who contributed to this report.

†Chair of the subcommittee that developed this report.

‡Consulting members who contributed to this report.

The committee would like to thank H. Aoude and F. Vossoughi for their contributions to this report.

This report provides a synopsis of the existing testing methodologies for the determination of mechanical properties of hardened fiber-reinforced concrete (FRC). This report applies to the mechanical properties of conventionally mixed and placed FRC, including fiber-reinforced self-consolidating concrete (FRSCC), or fiber-reinforced shotcrete (FRS) using steel, glass, polymeric, and natural fibers.

The objective is to enable manufacturers to characterize the mechanical properties of hardened FRC and encourage researchers and testing laboratories to adopt common and unified test methods to build a meaningful database of mechanical properties of hard-

ened FRC materials and products. Test results from the test procedures used in this report are not intended for the design of FRC structures, but to gain a better understanding of factors influencing the determination of their mechanical properties and of FRCs and FRC products.

Keywords: compressive strength; fiber pullout; fiber-reinforced concrete; flexural fatigue resistance; flexural strength; impact resistance; multiaxial behavior; shear and torsion; tensile strength; toughness.

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Report on Indirect Method to Obtain Stress-Strain Response of Fiber-Reinforced Concrete (FRC)

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Barzin Mobasher*, Chair
Neven Krstulovic-Opara*, Secretary

Clifford N. MacDonald, Membership Secretary

Corina-Maria Aldea
Emmanuel K. Attigbo
Nemkumar Bonthia
Joaquim Oliveira Barros*
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Arnon Bentur
Andrzej M. Brandt
James I. Daniel
Sidney Freedman
Melvyn A. Gattass
Henry J. Molloy
Antoine E. Naaman*Members of the subcommittee that authored this report.
†Chair of the task group that drafted this report.
‡Deceased.

Development of proper design procedures for fiber-reinforced concrete (FRC) materials requires use of material tensile and compressive stress strains that reflect the contribution of fibers to the post-cracking behavior. While uniaxial tension tests provide the most fundamental material properties, conducting closed-loop tension tests are difficult to accomplish; therefore, methods based on indirect measurement of tensile properties using flexural tests are typically used.

This report presents the methodologies that are used for data reduction and presentation of the flexural test results in terms of an equivalent tensile stress-strain response for FRC materials. Existing methods for estimating uniaxial tensile stress-strain response of strain-softening and hardening FRCs from flexural beam-test data are introduced. Different approaches applied to beam tests based on elastic equivalent, curve fitting, or back-calculation of flexural data are introduced. These are divided into two general categories: elastic equivalent approach or inverse analysis method. In the elastic equivalent approach, a summary of available test methods for various code agencies are presented.

Using back-calculation methods, tools based on the finite element method and analytical closed-form solutions are presented. An approach is presented that uses closed-form moment-curvature relationships and obtains load-deflection responses for a beam of three- or four-point loading. The method is used to obtain equivalent parametric tensile stress and strain relationships for a variety of FRC materials. The methods are compared against the available residual strength and also elastically equivalent residual strengths obtained by different specimen geometries.

Results for a range of FRC materials studied show the back-calculated post-peak residual tensile strength is approximately 30 to 37 percent of the elastically equivalent flexural residual strength for specimens with different fiber types and volume fractions.

Keywords: fiber-reinforced concrete; inverse analysis; tensile stress-strain diagram

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FRC 2014 Joint ACI-fib International Workshop
Fibre-reinforced concrete: From design to
structural applications, Montreal, Quebec

FRC 2018, Joint ACI-fib-Rilem International
Workshop, Lake Garda, Italy

Members of ACI 544 and *fib* TG 4.1 have been
involved in writing code-based specifications for
the design of FRC structural members.

Model code has been using Fibers for Structural
Design for several years now

fib Model Code 2010 includes specific sections
for design of FRC elements.

Bulletin 79



Fibre-reinforced concrete:
From design to structural applications
FRC 2014:ACI-fib International Workshop



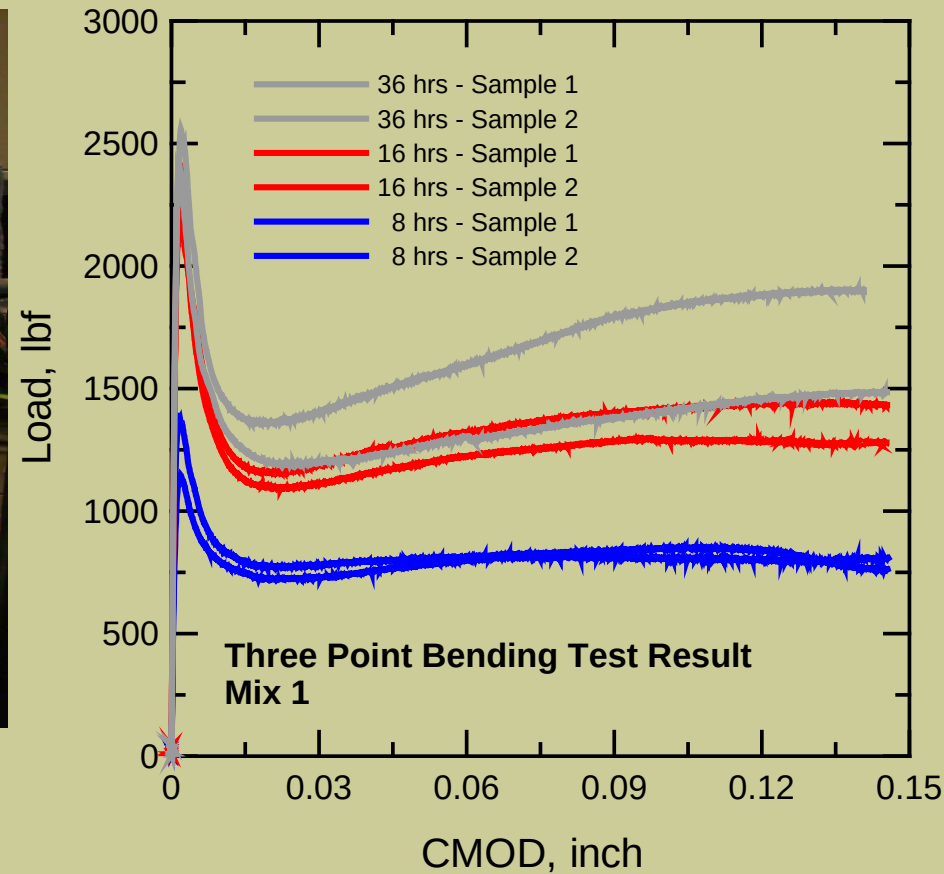
ACI SP-310
ACI-fib workshop proceedings

Model Code Provisions for use of FRC

- The design rules (for reinforced concrete – RC – and fibre-reinforced concrete – FRC) are present in the Model Code 2010 (2012).
- The safety factors of materials adopted are $\gamma_c = 1.5$ for concrete in compression, $\gamma_F = 1.5$ for FRC in tension, and $\gamma_s = 1.15$ for steel rebars.
- Moreover, fundamental and characteristic load combinations were considered at Ultimate Limit State (ULS) and Serviceability Limit State (SLS), respectively.
- The FRC used to exemplify the comparison in the design has a class 3c (Section 5.6.3 of MC 2010) which, as an example, can be obtained quite easily with a concrete C30/37 and an amount of steel fibres ranging between 30 and 40 kg/m³.

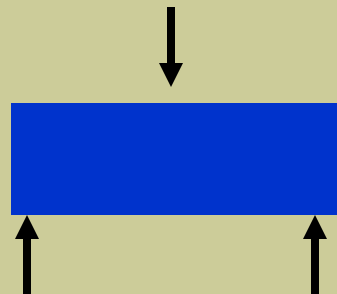
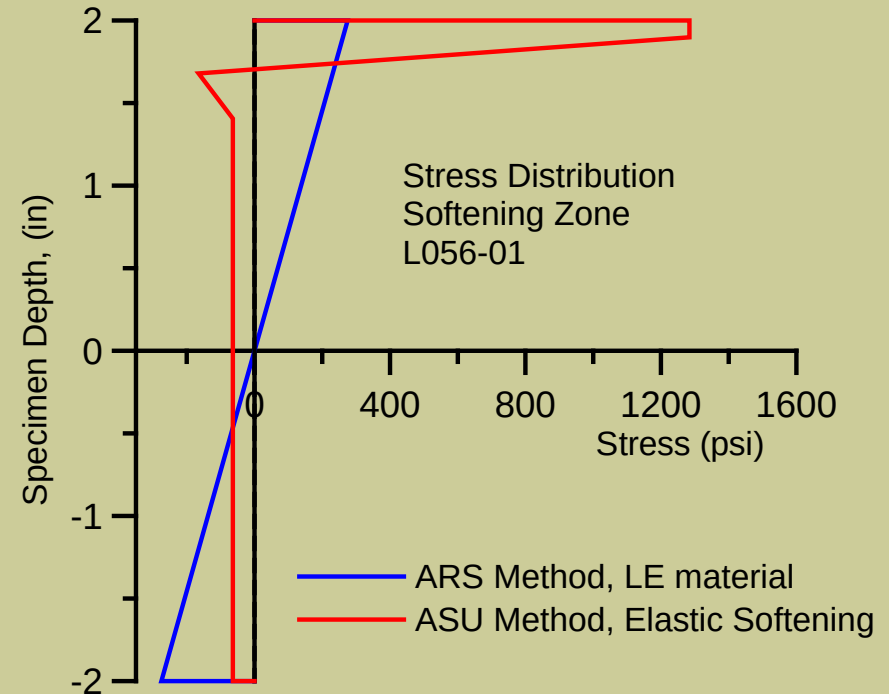
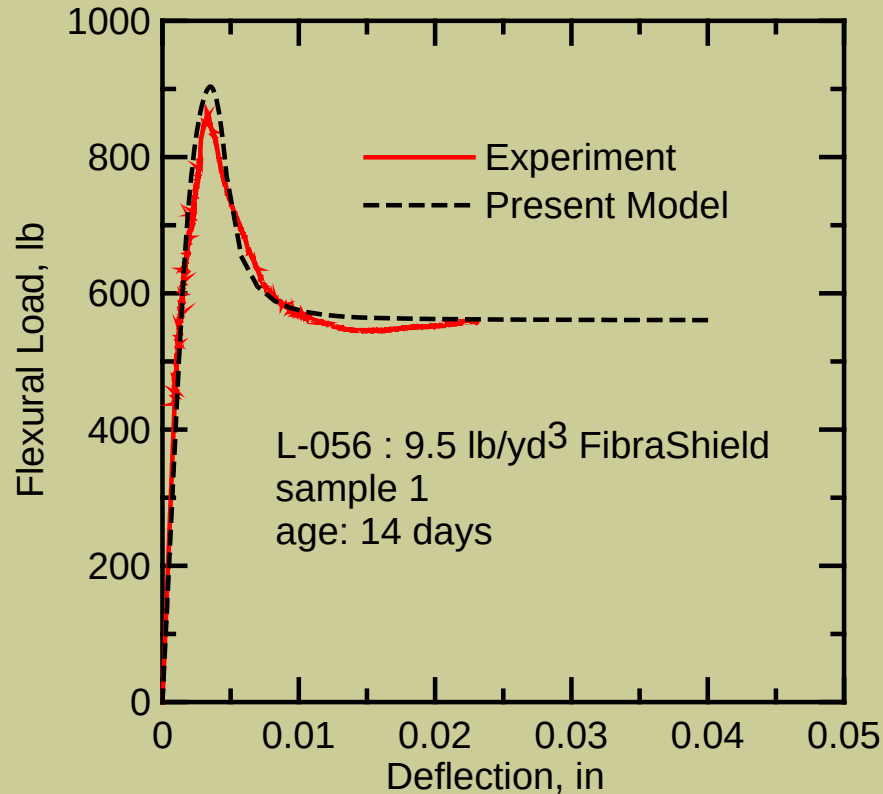
di Prisco, G., Plizzari, G., Vandewalle, L. (2009), Fibre reinforced concrete: new design perspectives, *Materials & Structures*, 42(9), 1261-1281.

Consideration of Residual strength Effect



10-12 lbs/yd³ of macro fibers

Stress distribution for a cracked material with Strain-Softening FRC

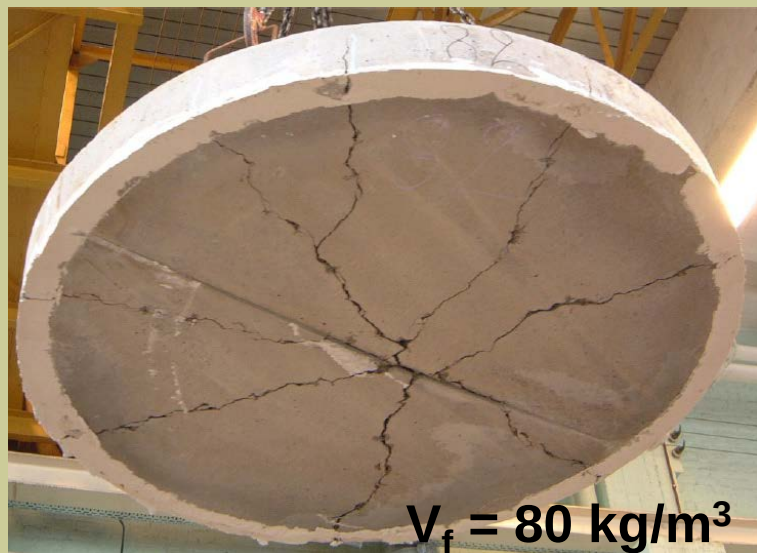


ASTM C1550-Round Panel 3P-support specimen



Round panel tests for evaluation of SFRC

- Test setup
 - displacement controlled, continuous support, center point load
- Dimensions
 - $D, t = 1500, 150$ mm
 - stroke diameter = 150 mm

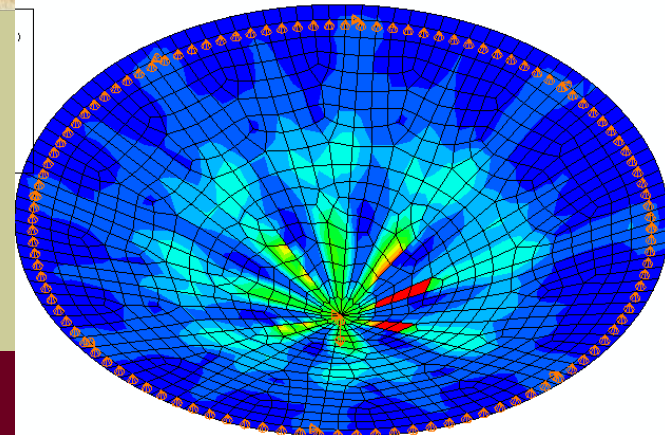
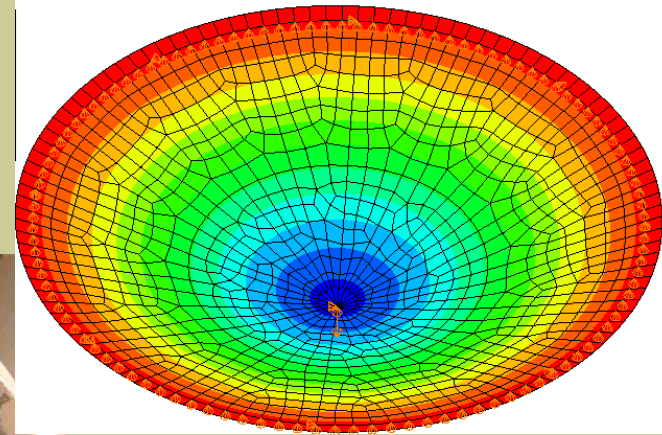
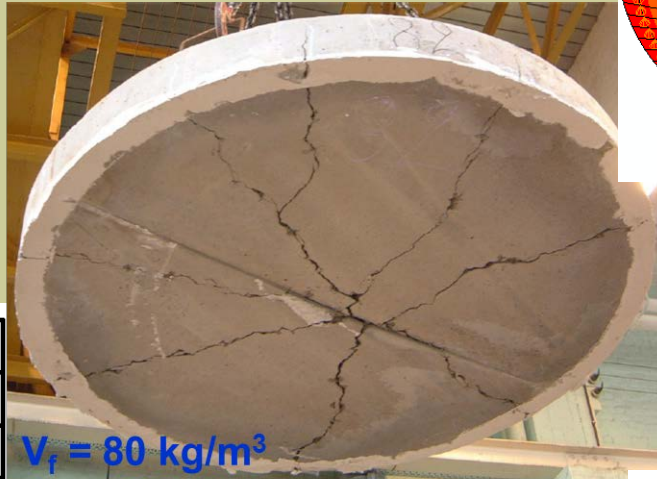
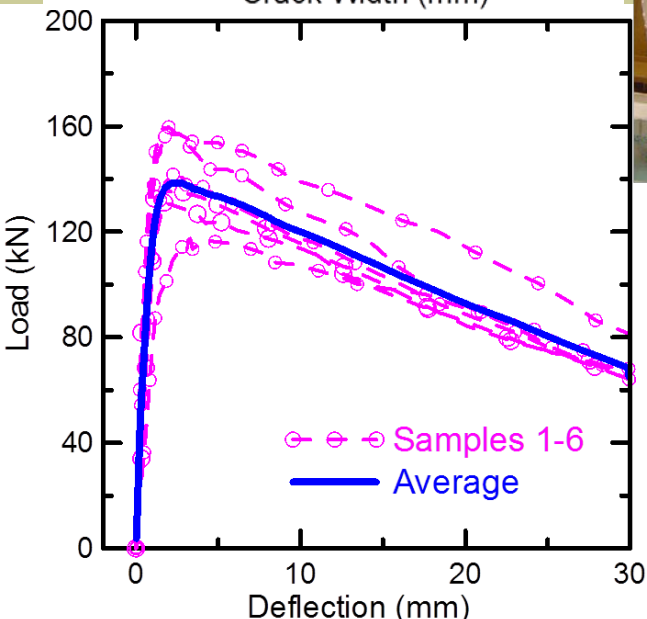
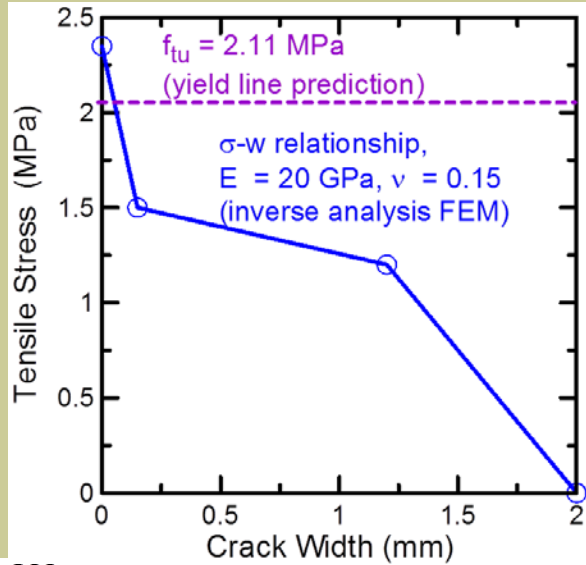


$V_f = 80 \text{ kg/m}^3$



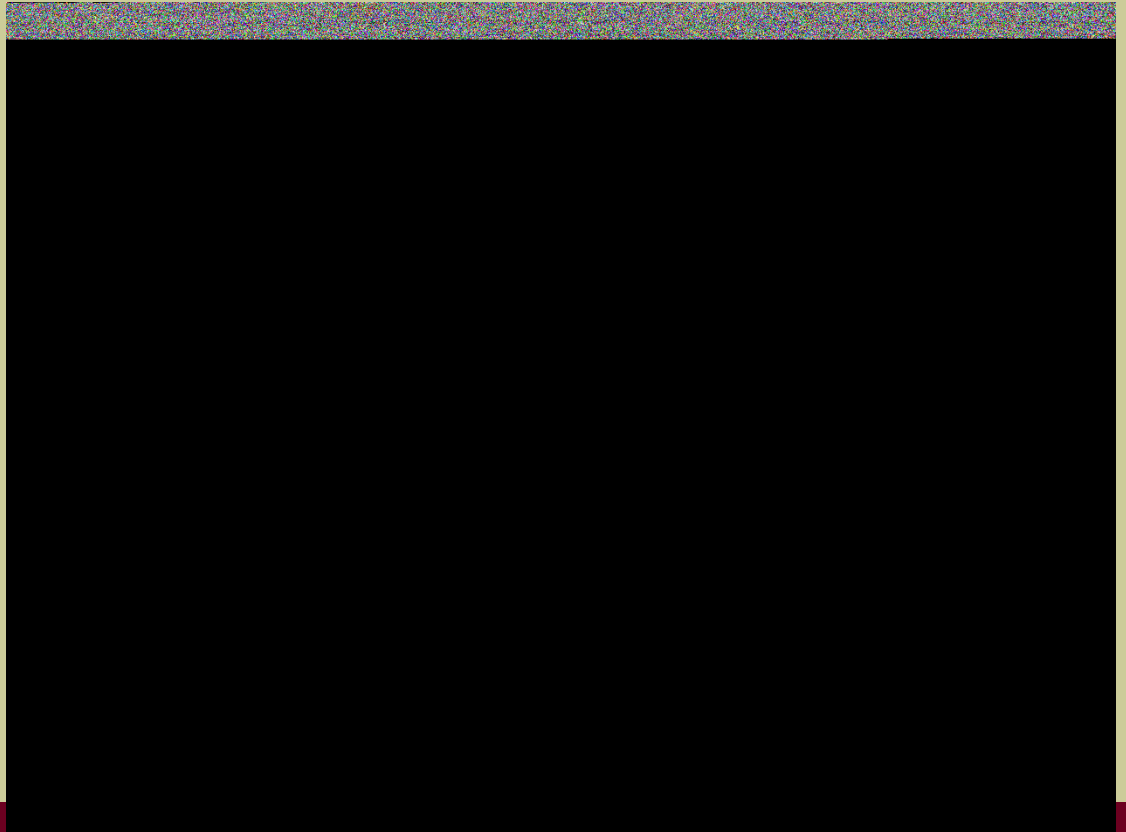
$V_f = 100 \text{ kg/m}^3$

Round Panel Continuous Support Specimens

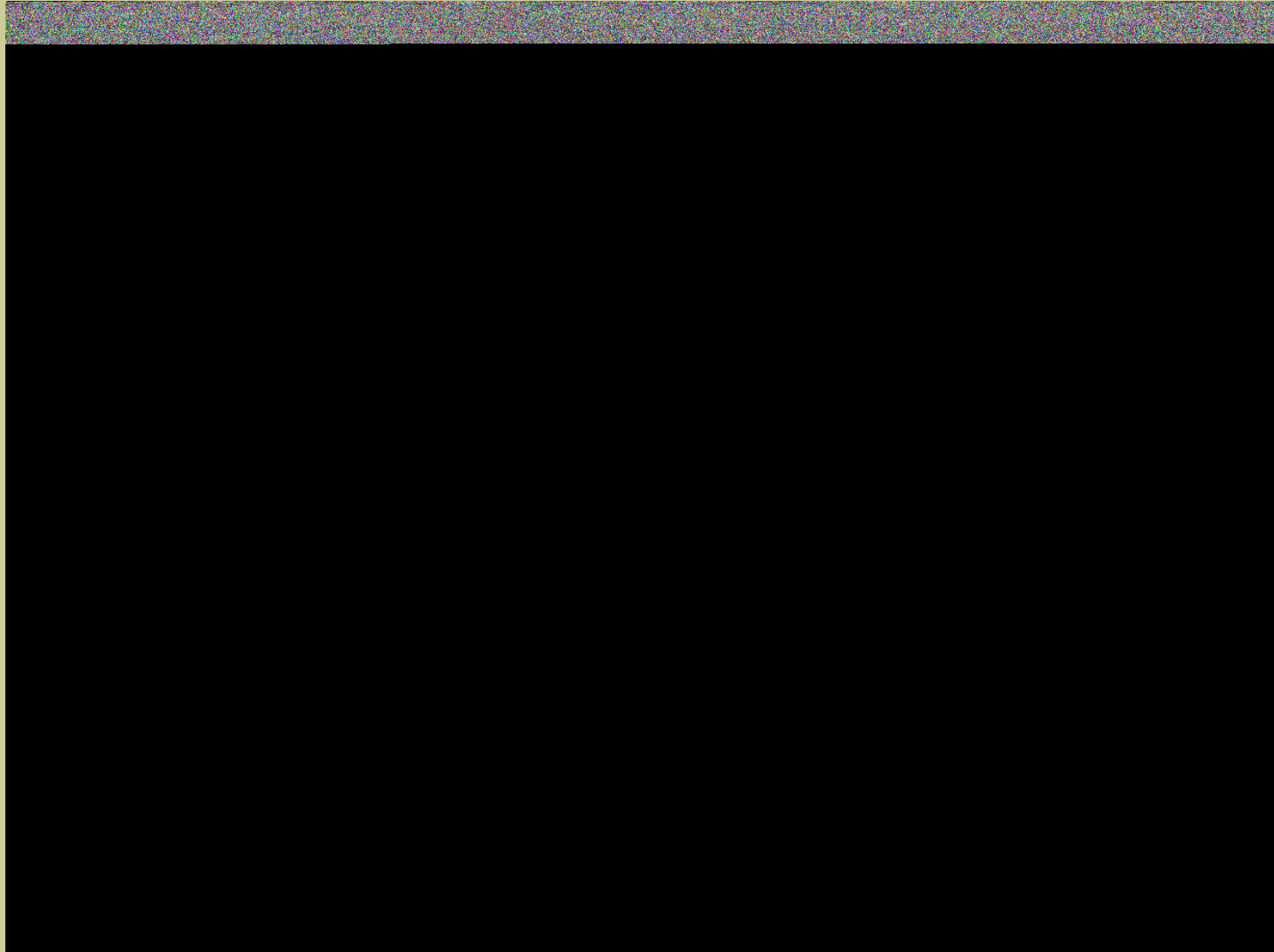


Advantages of using SFRC elevated slab: Pumping Reinforcement

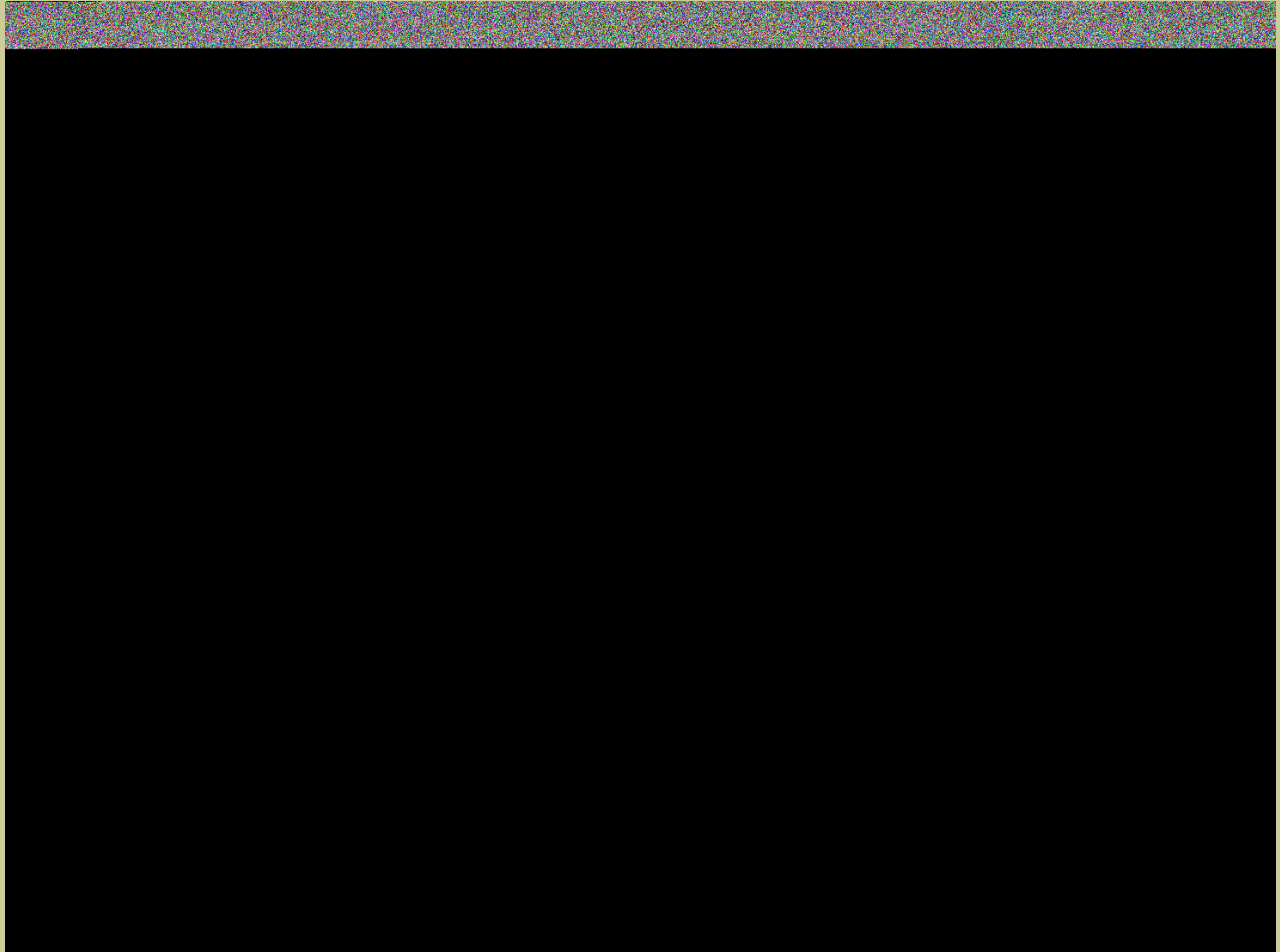
- Reduction in labor force and finishing personnel
- 30% cost saving vs. traditional methods
- Shrinkage cracking control
- Bay areas larger than traditional areas
- Detailing cost reduction



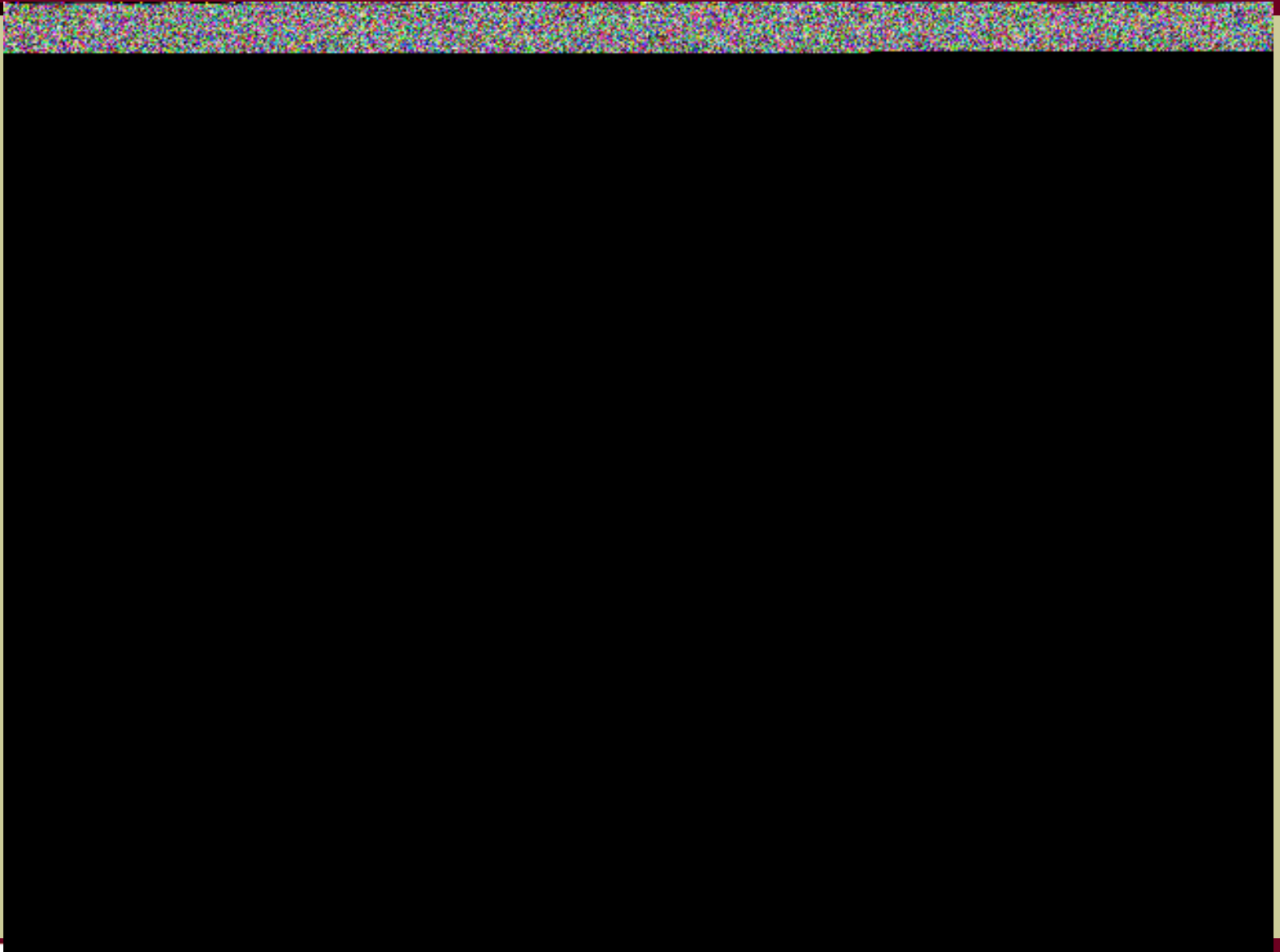
DittonNams-Superstore elevated slab Daugavspils, Latvia
25 ft span, 10inches slab,200psf UDL
165lbs/cu.yd Tabix 1.3/50 140ksi Steelfibers



**LKS struct.engineers headoffices, 5 floors 50000 sq.ft total
,Mondragon, Spain, 27ft span, 11 inches thickness, 140psf UDL
165lbs/cy.yd Tabix 1.3/50 140ksi steelfibers**



Several full scale tests, center span



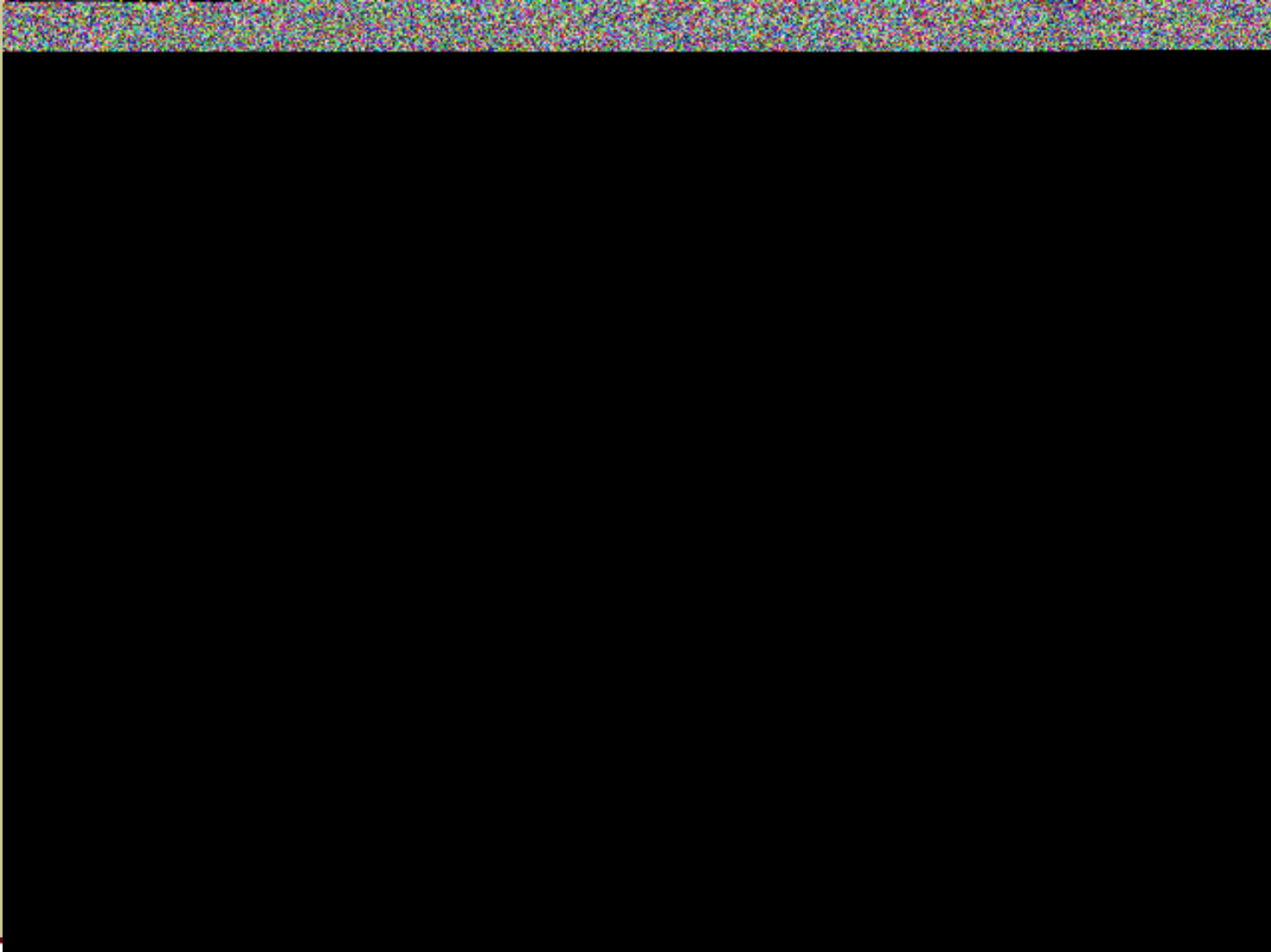
Tallinn, Estonia , Test site Service Load, 4kNm^2 udl, (83 psf)



Center Point loading Bissen Test



Corner span center point loading Test

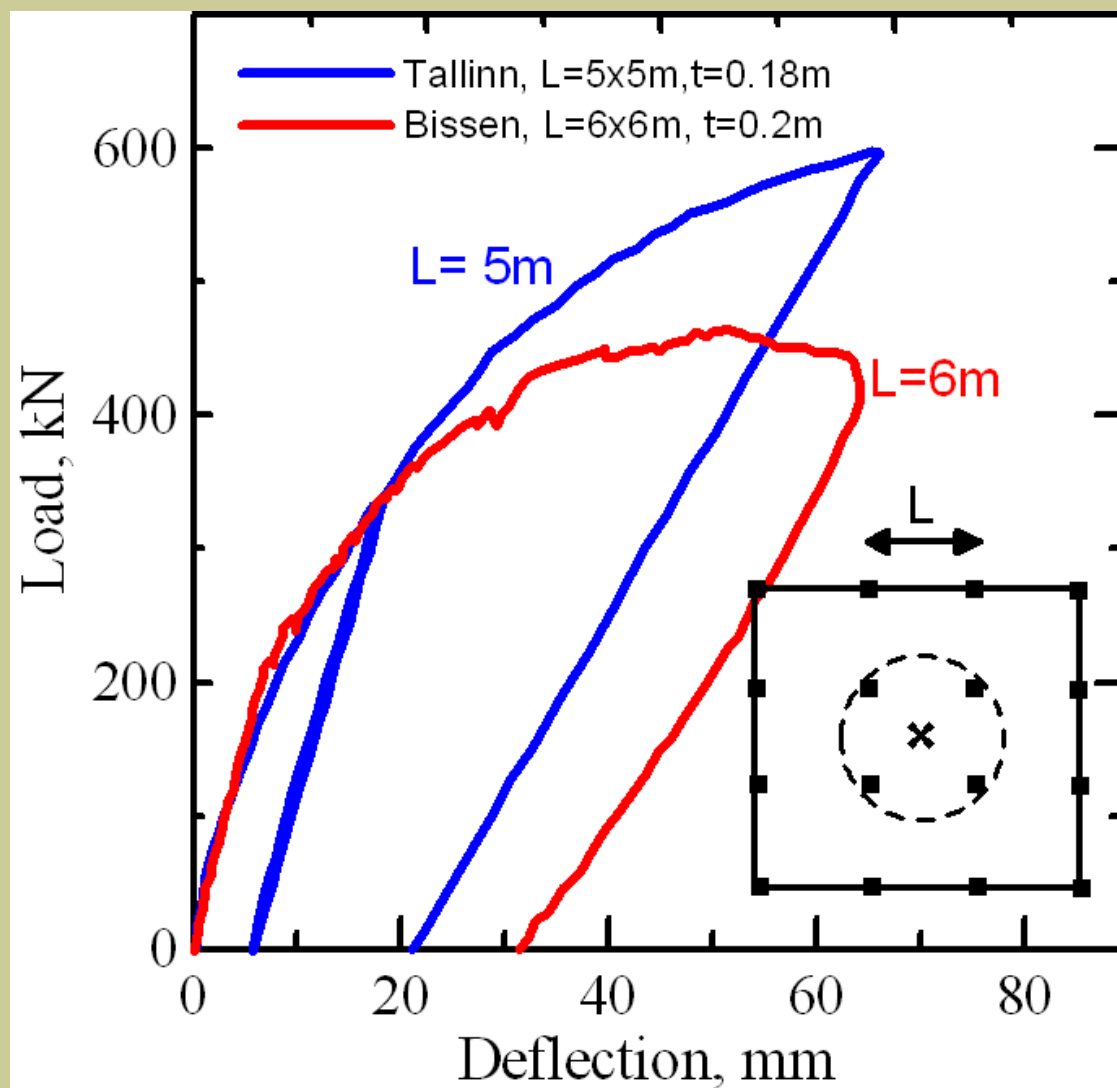


320kN 0.55 mm crack opening

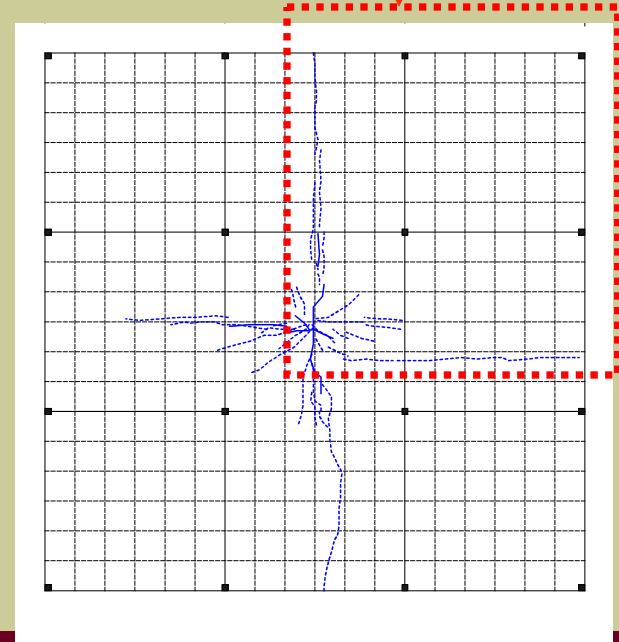
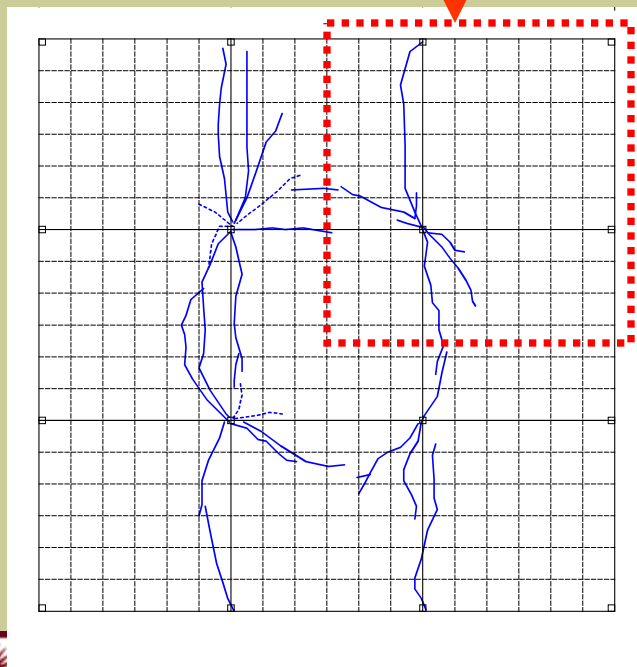
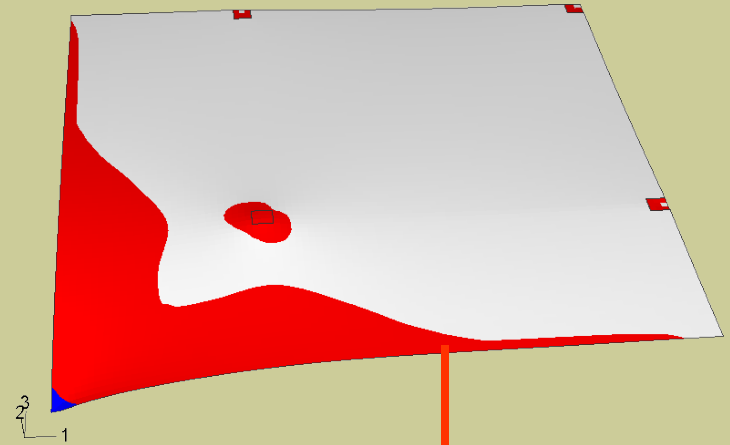
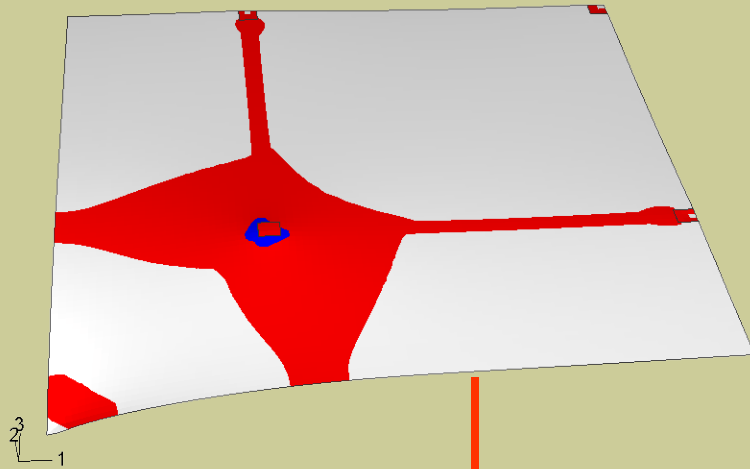
more than 3 x the first crack load!



Full scale Test Results



Modeling of Failure Mechanisms



Plastic analysis approach

Distributed load on a simply supported square slab. The work equations are derived as:

$$W_{int} = W_{ext}$$

Where the resultant N_R and rotation θ (from figure 9a) are:

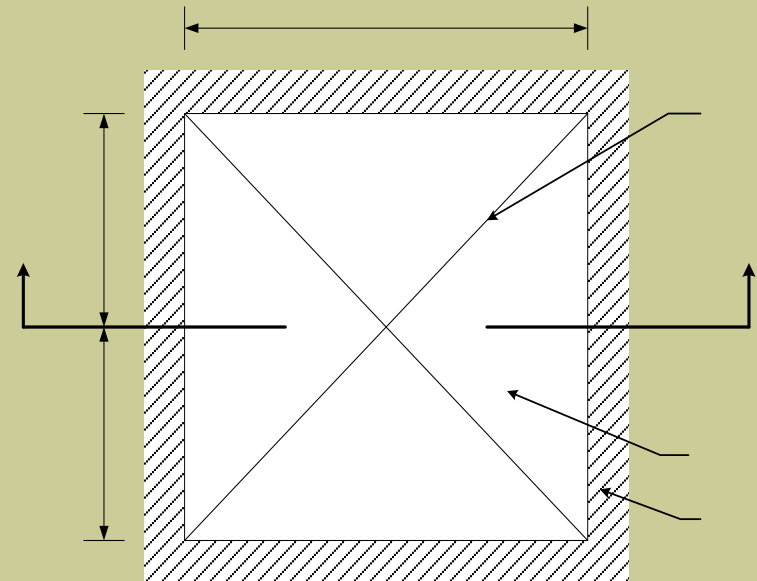
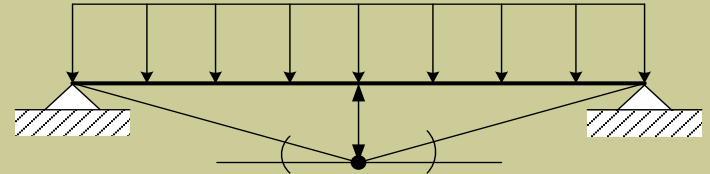
$$\sum (N_R \times \delta) = \sum (M \times L \times \theta)$$

$$N_R = q \times \left(\frac{L}{2}\right) \times \left(\frac{L}{2}\right) = \frac{qL^2}{4} \quad \theta = \frac{2\delta_{max}}{L}$$

For the four segments with an N_R acting at $1/3$ of δ_{max} :

$$4 \times \left(\frac{qL^2}{4}\right) \times \left(\frac{\delta_{max}}{3}\right) = 4 \times (M) \times (L) \times \left(\frac{\delta_{max}}{\frac{L}{2}}\right)$$

$$M_p = \frac{q_{ult} L^2}{24}$$

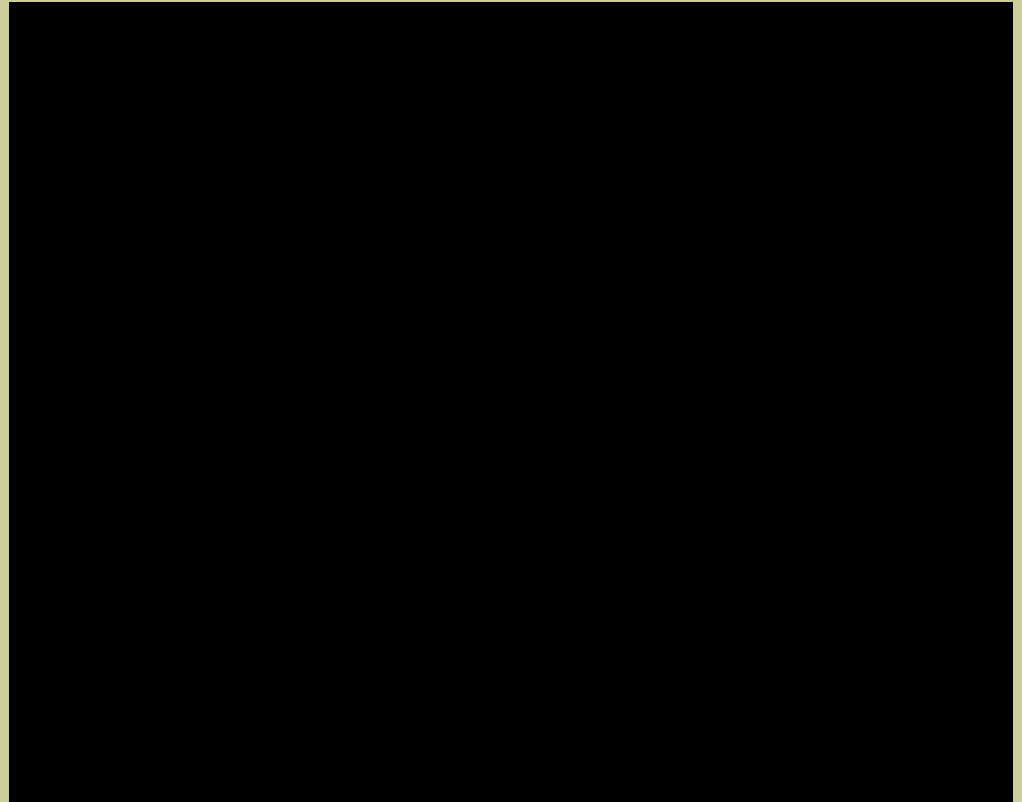


Use of Yield Line Analysis and Design procedures

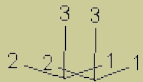
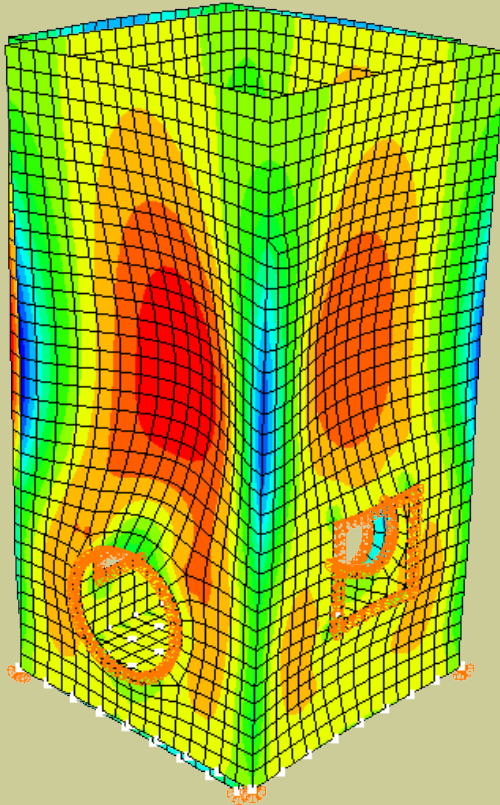
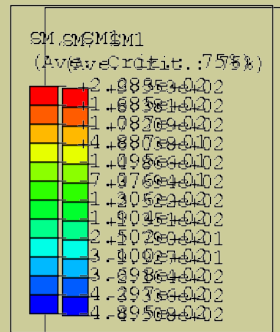
$$W_{int} = W_{ext}$$

$$2M_p\theta = P\delta = P\frac{L\theta}{2}$$

$$M_p = \frac{P_{ult}L}{4}$$



Analysis, Design and Installation of precast water tank panels

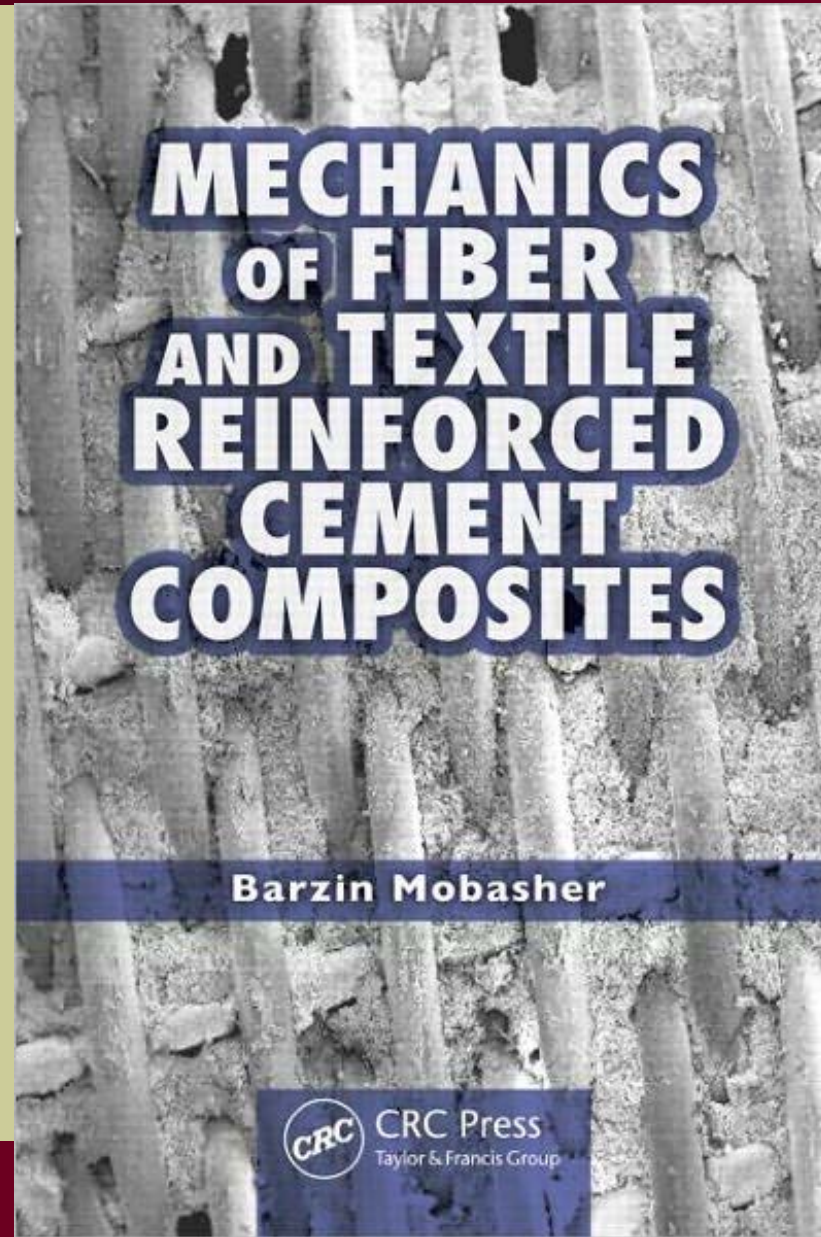


■ Load Case2:

- 1.4 Self weight +
1.7 Earth pressure +
1.7 Uniform pressure due to surcharge
- Moment in short span direction SM1

Mechanics of Fiber and Textile Reinforced Cement Composites

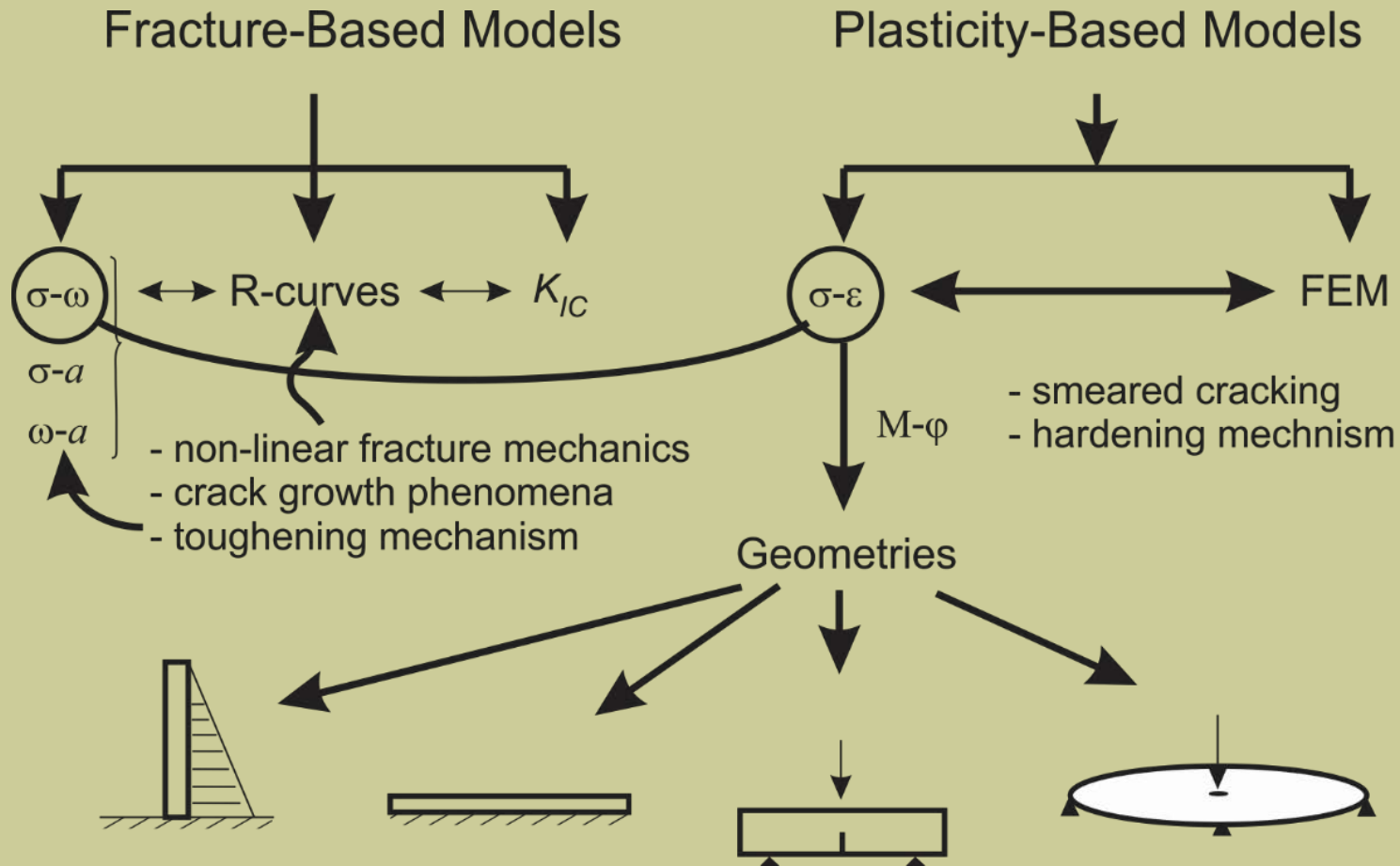
- CRC press, 2011
- E-mail: barzin@asu.edu
- <http://enpub.fulton.asu.edu/cement/>



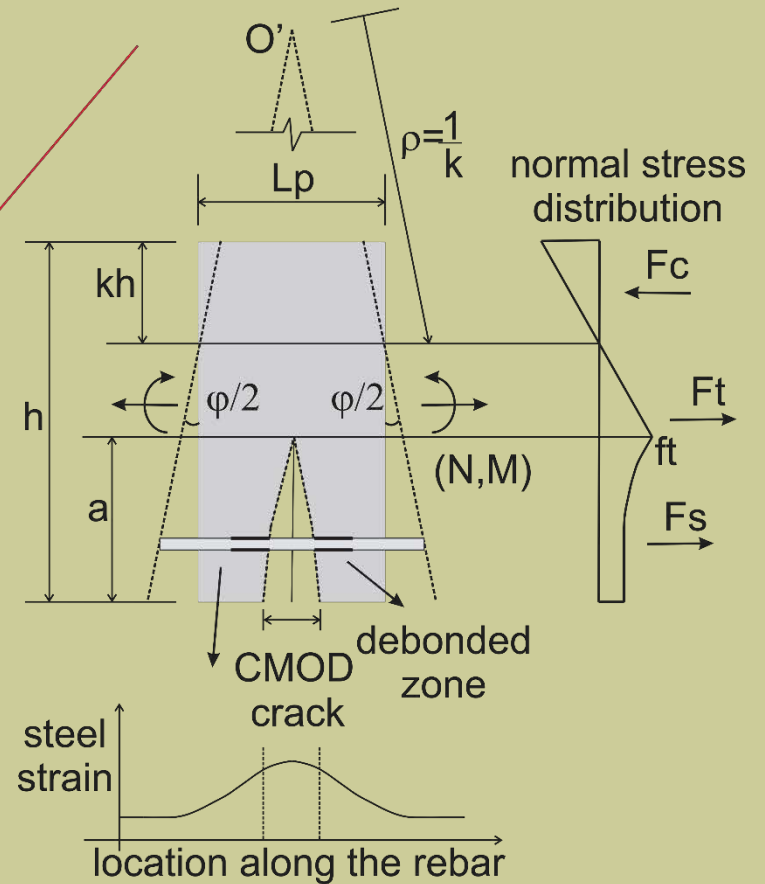
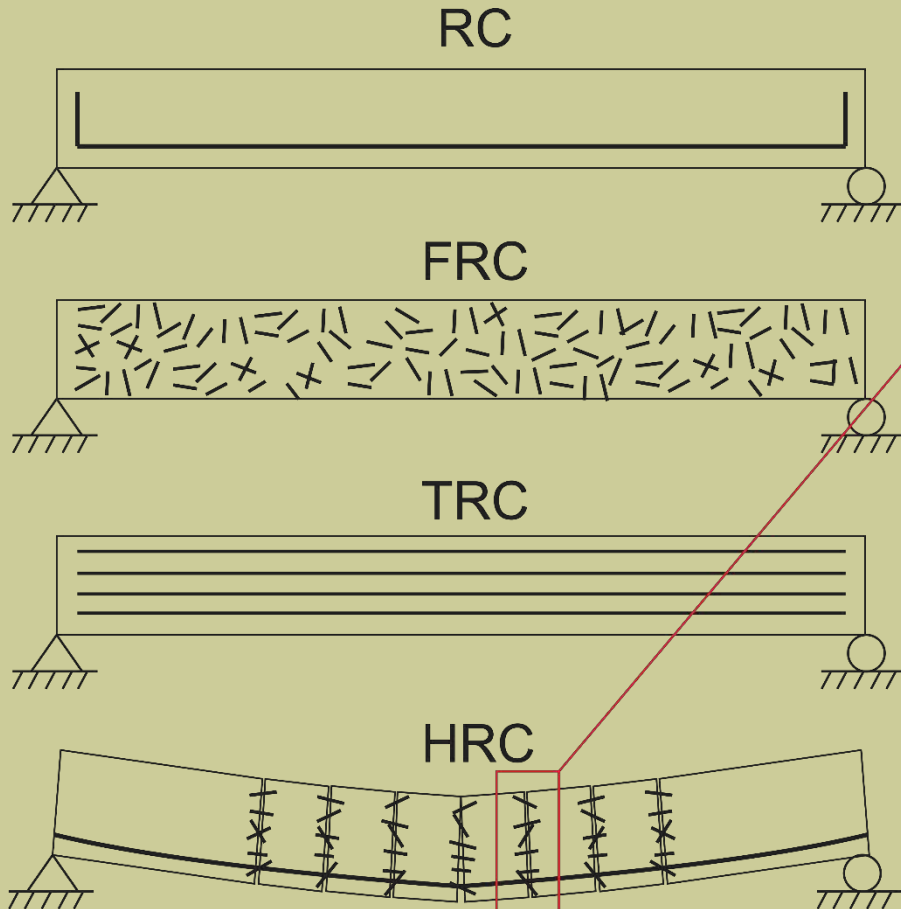
Conclusions

- New technologies and directions are clearly available in our future progression in Construction industry.
- Use of materials science based and mechanics based approaches to obtain better ways to characterize, model, analyze, and design.
- Better train, communicate, and follow through
 - Design of Materials and Structures for novel construction.
- Address specifications, quality, and long term performance.
- Fiber Reinforced Concrete can be effectively used as a technology enabler for our alternative energy generation, use and infrastructure development

Fracture and plasticity models



Localized hinge

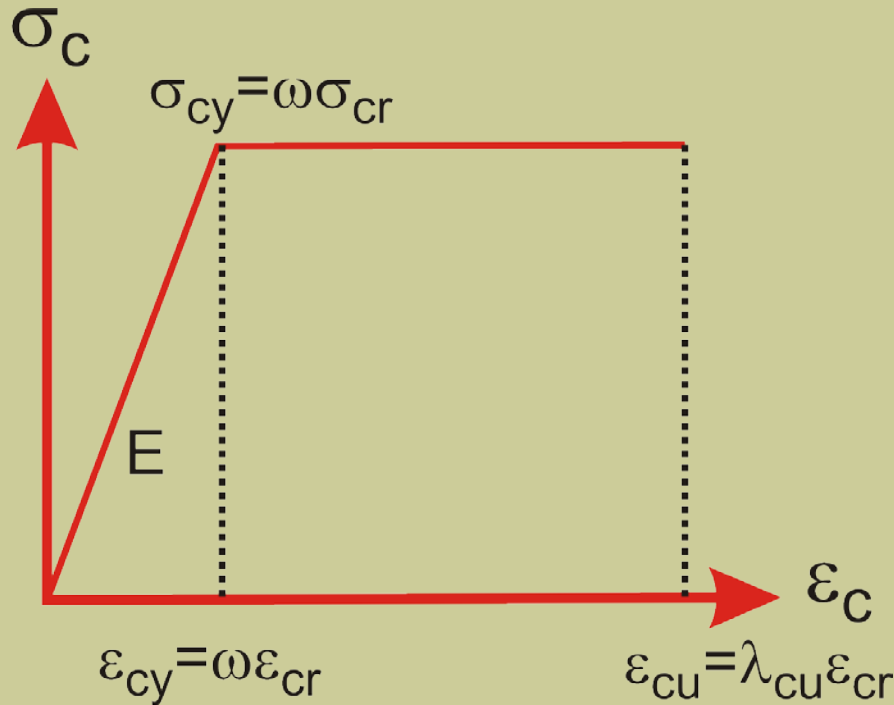


Modelling procedures

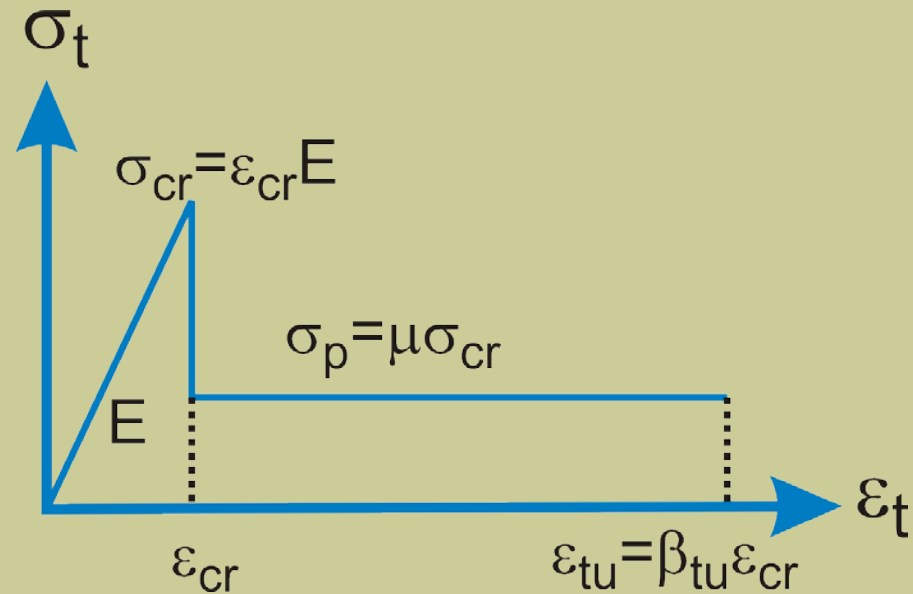
- Tension model
- Compression model
- Bond (interface) model
- Stress/strain distribution
- Moment-curvature relationship
- Normalized parameters
- Incorporations of equilibrium equations
- Moment and curvature distributions throughout the volume
- Integration, application of boundary conditions
- Data reduction and solutions for stress, strain, load, deflection distributions

Stress-Strain for Hardening and Softening FRC

- Material parameters are described as a multiple of the first cracking tensile strain (ε_{cr}) and tensile modulus (E)



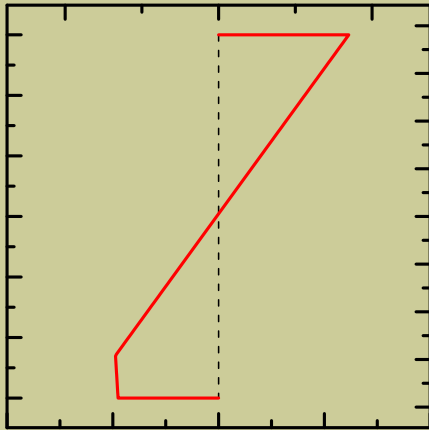
Compression model



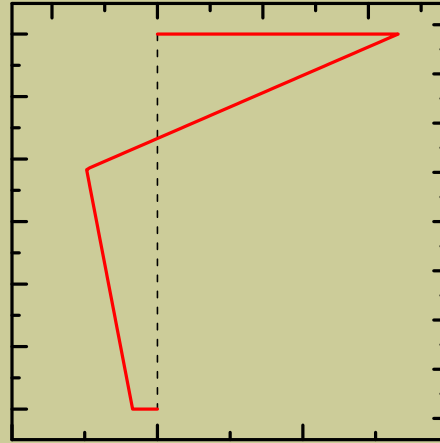
Tension model

Evolution of Stress Distribution Profile

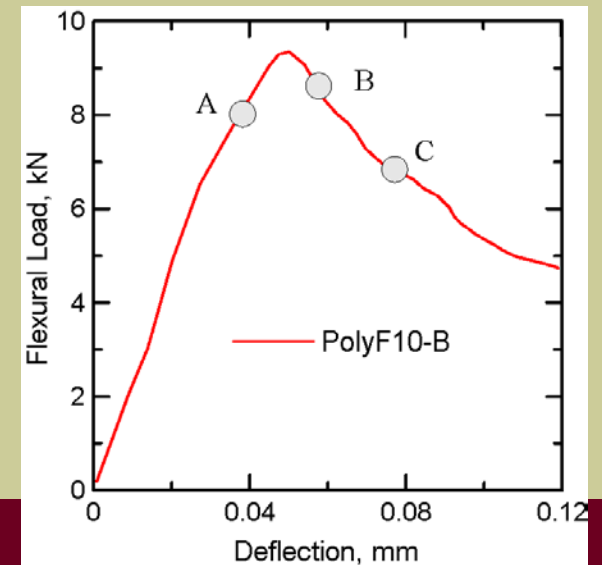
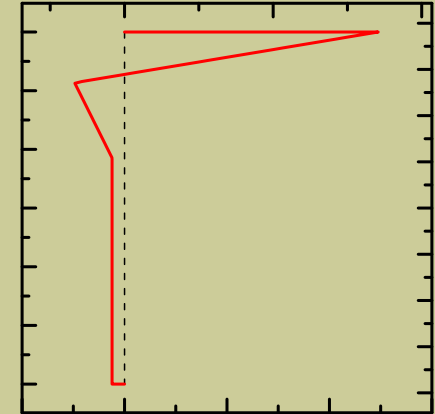
(A)



(B)

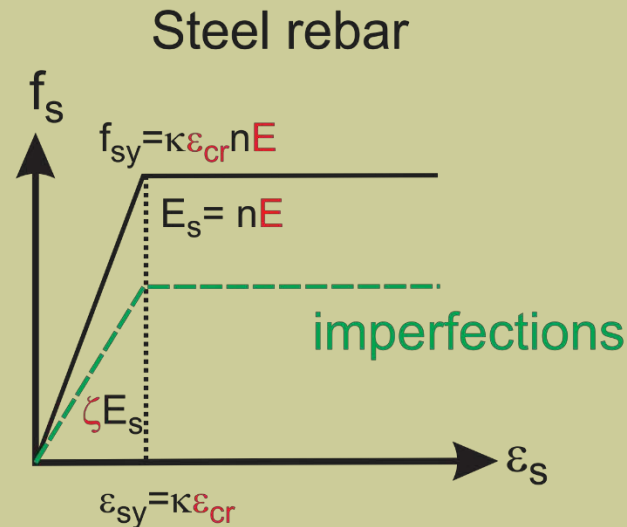
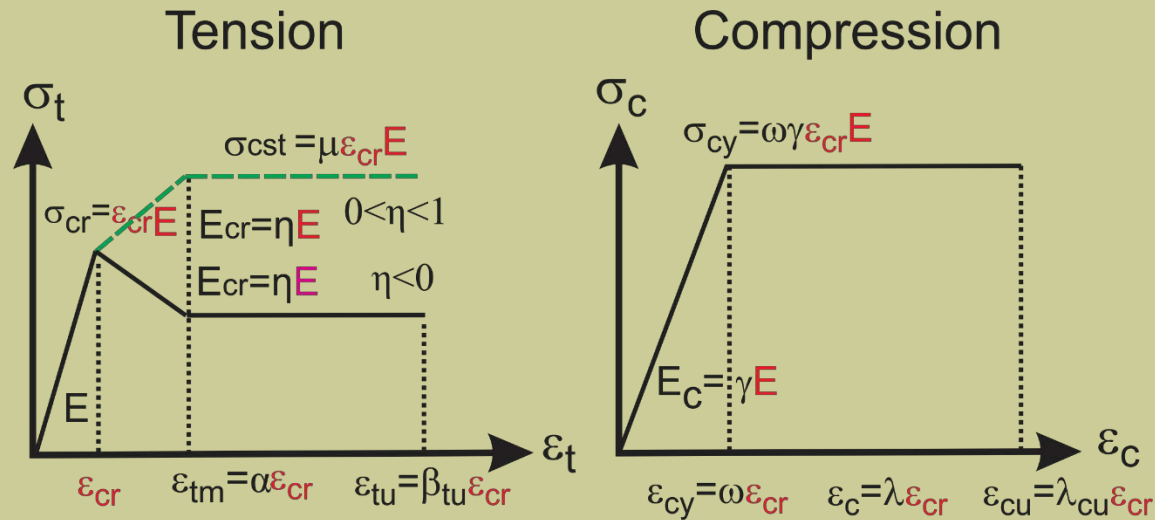


(C)

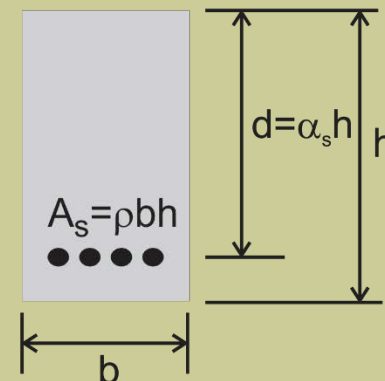


Material Models for Hybrid systems-

How much of the rebar may I take out and how much fiber to use for replacement?



Cross section



Derivation of Moment-Curvature Relationship

Incrementally impose

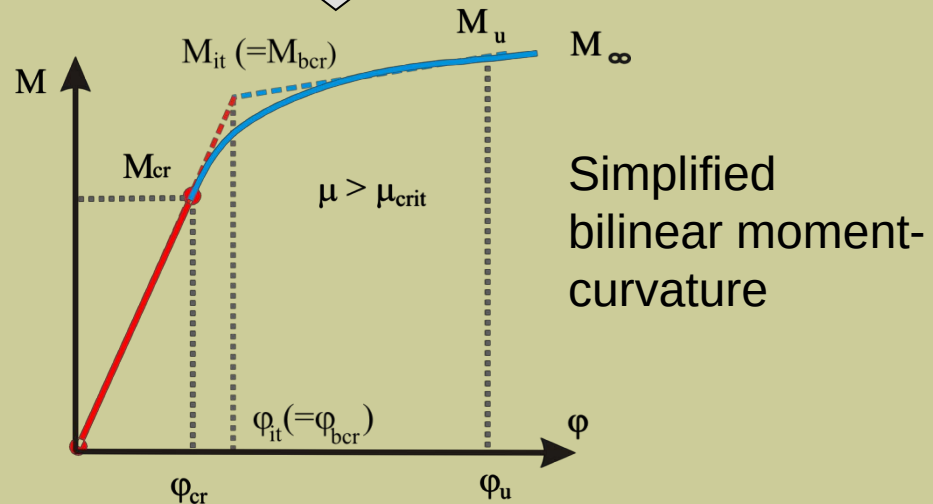
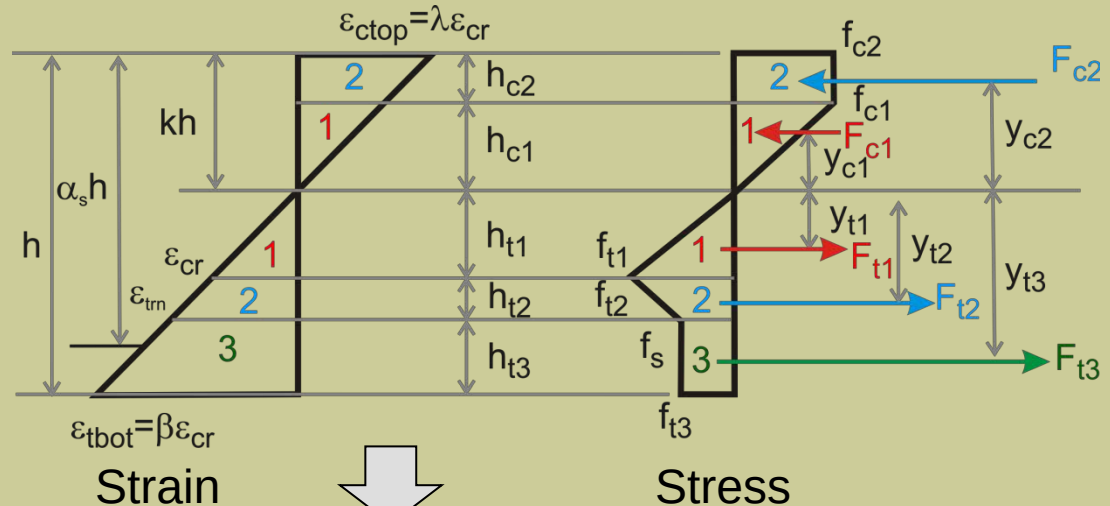
$$0 \leq \varepsilon_t \leq \varepsilon_{tu}$$

Strain Distribution
Stress Distribution

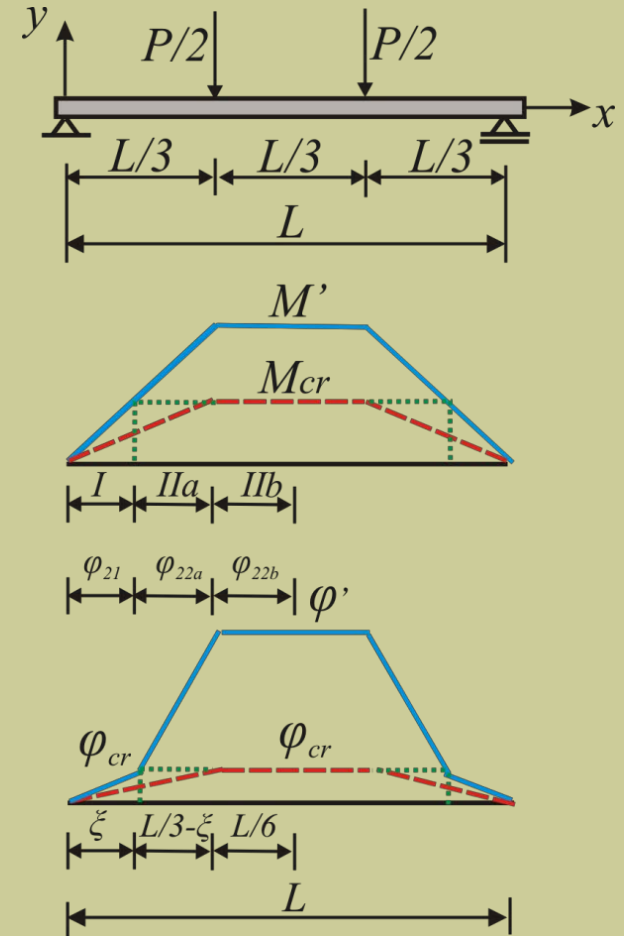
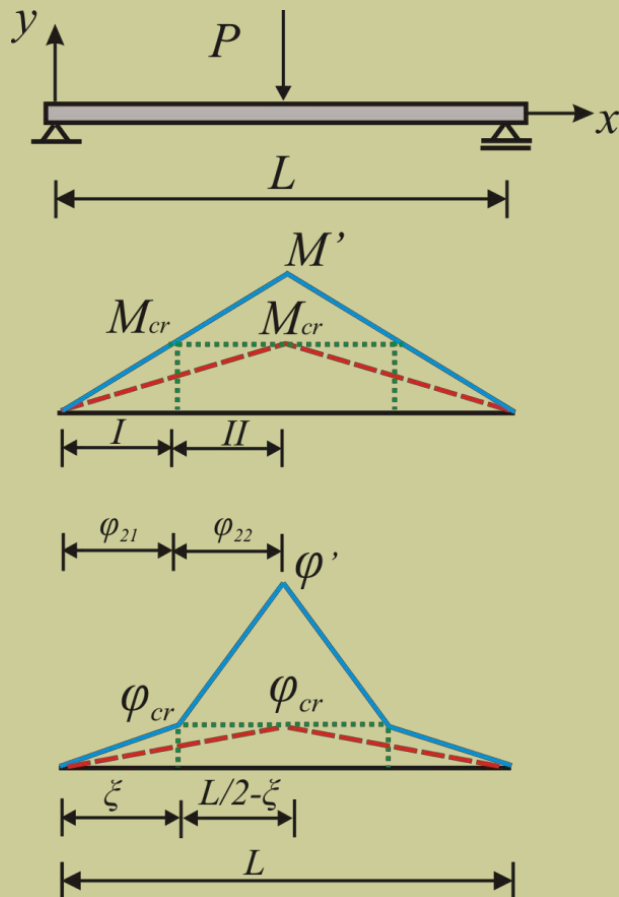
$\Sigma F = 0$, determine k

Moment: $M = \Sigma F_{ci}y_{ci} + \Sigma F_{ti}y_{ti}$
Curvature: $\phi = \varepsilon_c/kh$

Stage : $\lambda > \omega$, $\beta > \alpha$



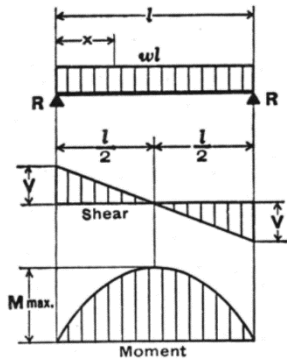
Moment-Curvature Distributions



- Use Static Equilibrium to get moment distribution
- Curvature distributions along the beam are generated based on simplified bilinear moment-curvature model

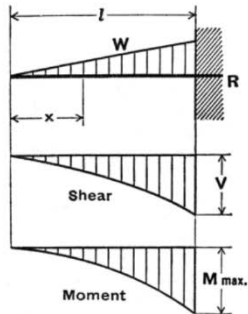
Linear elastic solutions-Steel Design Charts

1. SIMPLE BEAM—UNIFORMLY DISTRIBUTED LOAD



$$\begin{aligned}
 \text{Total Equiv. Uniform Load} & \dots = wl \\
 R = V & \dots = \frac{wl}{2} \\
 V_x & \dots = w \left(\frac{l}{2} - x \right) \\
 M_{\text{max. (at center)}} & \dots = \frac{wl^2}{8} \\
 M_x & \dots = \frac{wx}{2} (l - x) \\
 \Delta_{\text{max. (at center)}} & \dots = \frac{5wl^4}{384EI} \\
 \Delta_x & \dots = \frac{wx}{24EI} (l^3 - 2lx^2 + x^3)
 \end{aligned}$$

18. CANTILEVER BEAM—LOAD INCREASING UNIFORMLY TO FIXED END



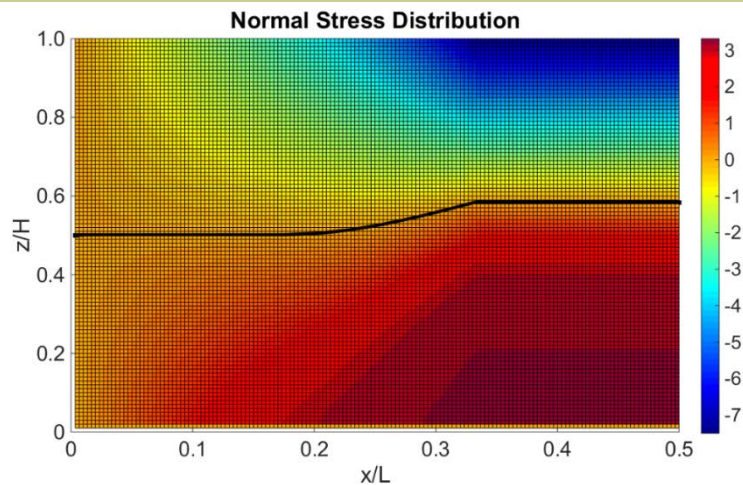
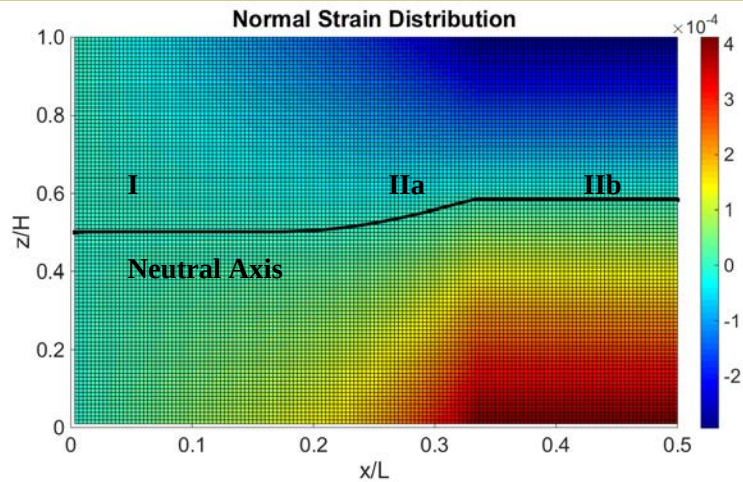
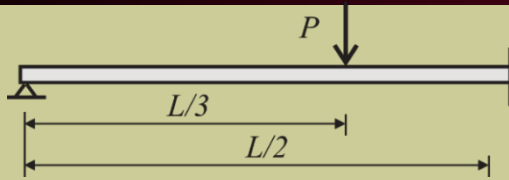
$$\begin{aligned}
 \text{Total Equiv. Uniform Load} & \dots = \frac{8}{3} W \\
 R = V & \dots = W \quad W = \frac{wl}{2} \\
 V_x & \dots = W \frac{x^2}{l^2} \\
 M_{\text{max. (at fixed end)}} & \dots = \frac{Wl}{3} \\
 M_x & \dots = \frac{Wx^3}{3l^2} \\
 \Delta_{\text{max. (at free end)}} & \dots = \frac{Wl^3}{15EI} \\
 \Delta_x & \dots = \frac{W}{60EI l^2} (x^5 - 5l^4x + 4l^5)
 \end{aligned}$$

Non-linear solutions

Beam Type	ξ	Region 1	Region 2	Curvature ϕ_{21}	Curvature ϕ_{22}	Deflection coefficient δ_{21}^*	Deflection coefficient δ_{22}^*
3PB	$\frac{L}{2m'}$	$0 < x < \xi$	$\xi < x < \frac{L}{2}$	$\frac{\phi_{cr} x}{\xi}$	$\phi_{cr} \left(\frac{2(q'-1)(x-\xi)+1}{L-2\xi} \right)$	$g_1 + g_2$	$g_3(g_4 + g_5 + g_6 + g_7)$
4PB	$\frac{L}{3m'}$	$0 < x < \xi$	$\xi < x < \frac{L}{3}$	$\frac{\phi_{cr} x}{\xi}$	$\phi_{cr} \frac{3(q'x - q'\xi - x) + L}{L - 3\xi}$	$f_1(f_2 + f_3)$	$f_4(f_5 + f_6 + f_7)$
			$\frac{L}{3} < x < \frac{L}{2}$		$q' \phi_{cr}$		$f_8 + f_9$
SS-Uniform loading	$\frac{L}{2} \left(1 - \sqrt{1 - \frac{1}{m'}} \right)$	$0 < x < \xi$	$\xi < x < \frac{L}{2}$	$-\frac{\phi_{cr}}{\xi(L-\xi)}(x^2 - Lx)$	$\frac{4\phi_{cr}}{(L-2\xi)^2} \left(\frac{xq'L - q'x^2 - xL + x^2}{- \xi q'L + q'\xi^2 + \frac{L^2}{4}} \right)$	j_1	$j_2(j_3 + j_4 + j_5 + j_6 + j_7)$
SS-Moment at the middle	$\frac{L}{2m'}$	$0 < x < \xi$	$\xi < x < \frac{L}{2}$	$\frac{\phi_{cr} x}{\xi}$	$\phi_{cr} \left(\frac{2(q'-1)(x-\xi)+1}{L-2\xi} \right)$	i_1	i_2
C-Uniform loading	$L \left(1 - \sqrt{\frac{1}{m'}} \right)$	$\xi < x < L$	$0 < x < \xi$	$\frac{\phi_{cr}(-x^2 + 2Lx - L^2)}{L^2 - 2L\xi + \xi^2}$	$\phi_{cr} \left(\frac{(q'-1)(x^2 - 2Lx)}{(\xi^2 - 2L\xi)} - q' \right)$	$n_1(n_2 + n_3 + n_4)$	n_5
C-Load at the end	$\xi = (1 - \frac{1}{m'})L$	$\xi < x < L$	$0 < x < \xi$	$\frac{\phi_{cr}(x-L)}{L-\xi}$	$\phi_{cr} \left(\frac{(q'-1)(x-\xi)}{\xi} - 1 \right)$	u_1	u_2
C-Load at any point	$\xi = (1 - \frac{1}{m'})S$	$\xi < x < S$	$0 < x < \xi$	$\frac{\phi_{cr}(x-S)}{S-\xi}$	$\phi_{cr} \left(\frac{(q'-1)(x-\xi)}{\xi} - 1 \right)$	$c_1(c_2 + c_3 + c_4)$	$c_9 + c_{10}$
		$S < x < L$		0		$c_5(c_6 + c_7 + c_8)$	

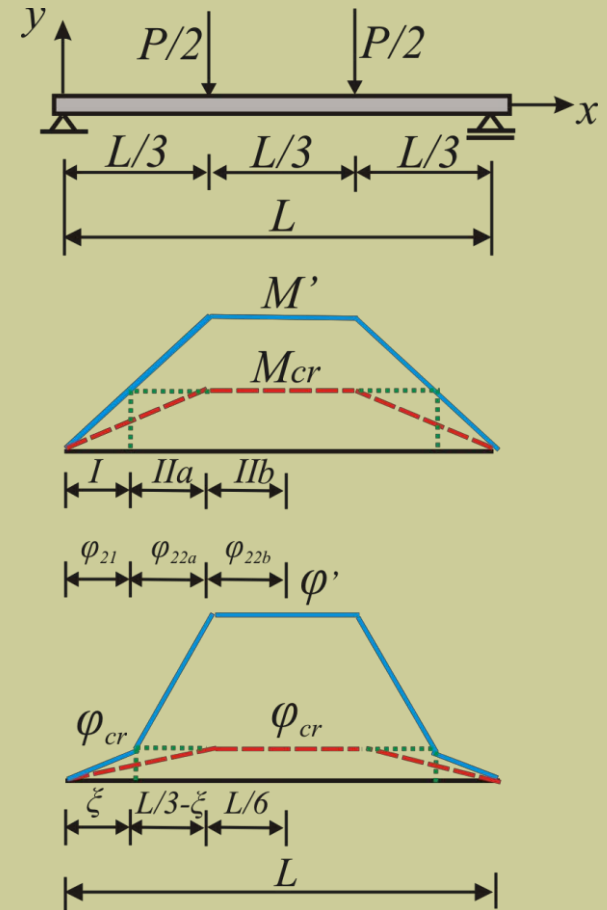
- 3PB, 4PB – 3, 4 point bending
- SS – simply supported
- C - cantilever

2D Contour – Distribution loading

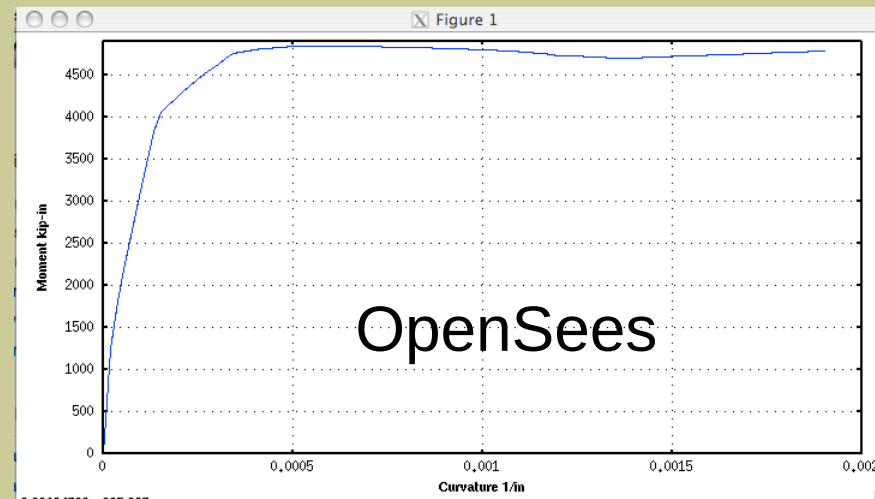
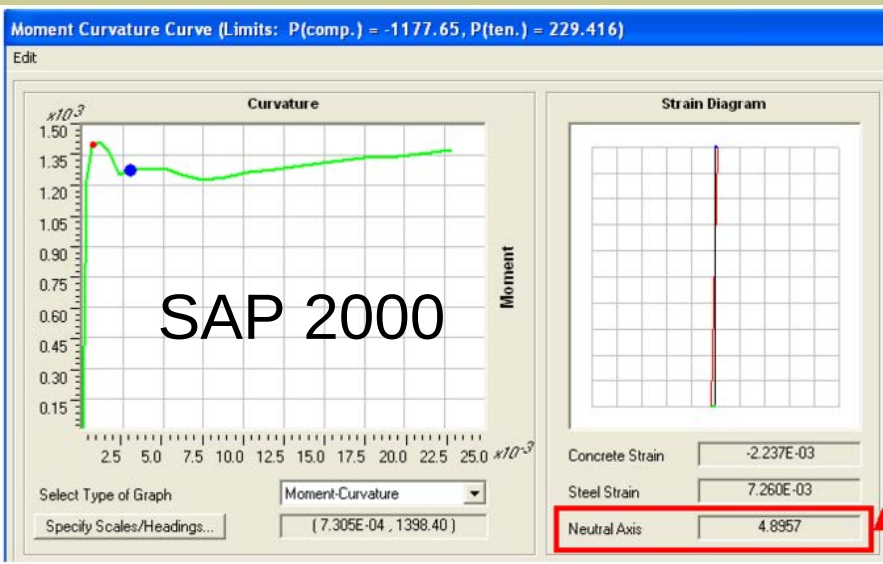
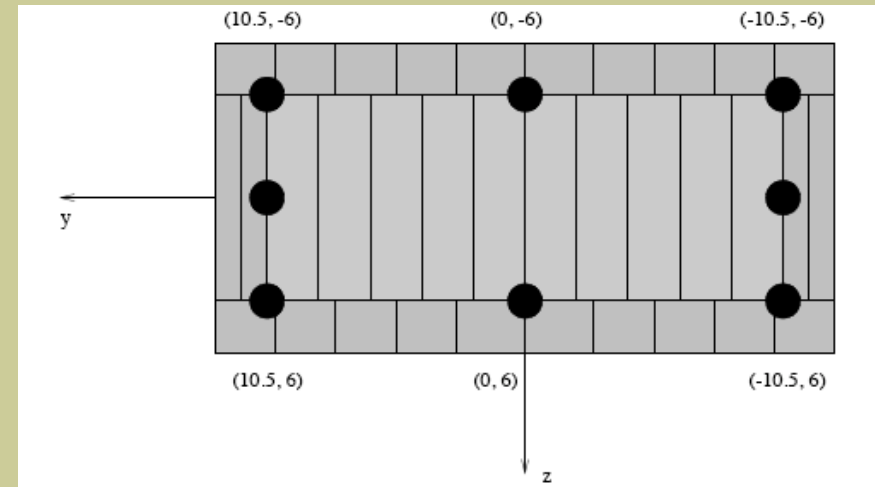
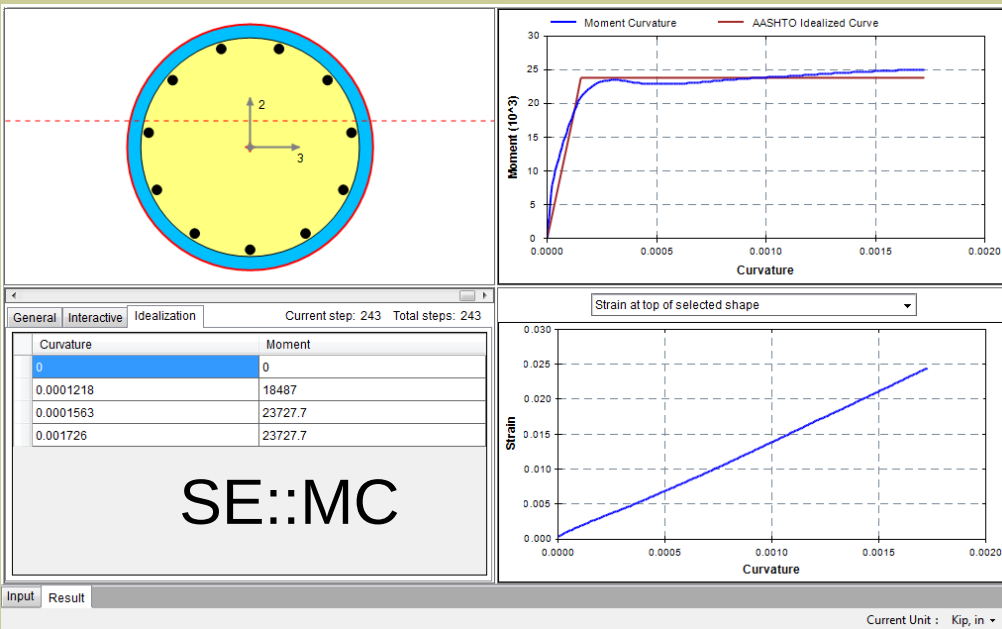


Normal strain distribution

Normal stress distribution



External software for moment-curvature analysis



Experimental Verification

- Three point bending tests
- Six full scales FRC beams
- Fiber content : 50kg/m³ and 75kg/m³

Geometry of specimens

Beams	H500	H1000	H1500
Height (mm)	500	1000	1500
Effective depth (mm)	440	940	1440
Total length (mm)	3000	5900	9000
Span(mm)	2640	5640	8640
Width(mm)	250	250	250
Reinforcement longitudinal bars	8-φ14	8-φ20	8-φ24

Parameters for current Model

Beam Type	Fiber Content kg/m ³	E	Simulated		
			E	ε _{cr}	η
		MPa	MPa	μstr	
H500	50	30800	30800	146	0.24
	75	32100	32000	110	0.26
H1000	50	30800	29500	125	0.22
	75	32100	32100	120	0.26
H1500	50	30800	29500	128	0.23
	75	32100	32100	117	0.26